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Title:

Using Machine Learning to Predict Customer Churn in E-Commerce: A Comparative Study with Emphasis on Business Interpretability

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Abstract

Customer churn has become a major challenge in the e-commerce industry because losing existing customers can negatively affect profitability and long-term business sustainability. Therefore, this study aims to develop and evaluate machine learning models for predicting customer churn in e-commerce with an emphasis on both predictive performance and business interpretability. The study utilised an E-commerce Customer Churn dataset consisting of 6,000 customer records with 15 input variables and one target variable indicating churn status. Several data preprocessing techniques were applied, including data cleaning, label encoding, data splitting, outlier detection and capping using the Interquartile Range (IQR) method, feature selection, feature scaling, and Synthetic Minority Oversampling Technique (SMOTE) to address class imbalance. Six supervised machine learning models were developed and compared, namely Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and XGBoost. The models were evaluated using confusion matrix, accuracy, precision, recall, F1-score, and Area Under the Receiver Operating Characteristic Curve (AUC-ROC). The findings revealed that Random Forest achieved the best overall performance balance, securing a classification accuracy of 96.2%, a precision of 0.843, a recall of 0.925, a peak F1-score of 0.882, and an outstanding AUC score of 0.992. In addition, SHapley Additive exPlanations (SHAP) analysis was applied to improve model interpretability by identifying the most influential variables affecting churn prediction. The results showed that engagement score and days since last purchase were the most significant factors contributing to customer churn. Overall, this study demonstrates that combining machine learning with explainable artificial intelligence techniques can support accurate, transparent, and data-driven decision-making for customer retention strategies in the e-commerce industry.

Keywords: Customer Churn Prediction, E-commerce, Machine Learning, Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), XGBoost, SHAP, Predictive Analytics, Customer Retention

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1.0 Introduction

1.1 Background of the Study

E-commerce, also known as electronic commerce, refers to the buying and selling of products and services through the internet and other digital platforms. It involves the use of information technology, including electronic data interchange, online payment systems, and web-based platforms to facilitate business transactions between firms and consumers (Jain et al., 2021). Through e-commerce, businesses can distribute products, conduct marketing activities, and provide services efficiently across geographical boundaries. The rapid growth of e-commerce has significantly transformed traditional business operations and contributed to economic development by enhancing market accessibility and global integration. In addition, the increasing reliance on digital transactions has generated large volumes of customer data, creating opportunities for businesses to better understand consumer behavior and improve decision-making processes (Kedah, 2023).

Customer churn refers to the likelihood that customers will stop doing business with a company within a specific period. It has become a major challenge for firms worldwide, leading many industries such as financial services, telecommunications, and airlines to adopt data mining techniques for churn analysis. However, limited research has focused on e-commerce churn due to its virtual nature, where interactions between companies and customers occur without face-to-face contact (Yu et al., 2011). This makes it more challenging for businesses to clearly identify when and why customers stop using online e-commerce platforms. In this context, customer data such as transactional and behavioural records, which capture past interactions and purchasing patterns over time are crucial for comprehending consumer behaviour and supporting with analytical decision-making (Baghla & Gupta, 2022; Yu et al., 2011).

Customer churn prediction techniques have been widely applied to identify customers who are likely to discontinue their relationship with a company, enabling firms to design more effective retention strategies. Retaining existing customers is essential because it helps prevent performance decline and reduces the cost associated with acquiring new customers (Xiahou & Harada, 2022). Customer churn prediction involves building a predictive model that assigns churn probabilities based on past customer behaviour and ranks customers from those most likely to leave to those least likely to leave. This enables businesses to identify high-risk customers and implement targeted retention strategies, while also tailoring appropriate approaches for non-churners. High-risk customers can then be approached with customised interventions and incentives to encourage continued engagement with the company (Gordini & Veglio, 2017).

In today's fast-paced digital environment, many businesses have increasingly adopted machine learning techniques to predict customer churn in e-commerce settings. Additionally, by extracting different kinds of data to enable churn prediction, e-commerce business managers can use big data and cloud computing to analyse and model consumer behaviour (Xiahou & Harada, 2022). Several studies have demonstrated that machine learning techniques significantly improve the accuracy and effectiveness of customer churn prediction by capturing complex patterns in customer behaviour (Baghla & Gupta, 2022; Gordini & Veglio, 2017; Xiahou & Harada, 2022; Yu et al., 2011).

Artificial intelligence (AI) refers to computer systems designed to simulate human intelligence, including learning, reasoning, and decision-making. Machine learning is a subset of AI that enables systems to learn from data without being explicitly programmed. Instead of relying on fixed instructions, machine learning models analyze large datasets to identify patterns and relationships. These patterns are then used to make predictions or decisions on new, unseen data, and the system gradually improves its accuracy as more data becomes available (Karimi et al., 2024). There are four main types of machine learning, namely supervised learning, unsupervised learning, semi-supervised learning, and reinforcement learning (Alnuaimi & Albaldawi, 2024). However, this study focuses on supervised learning because it is the most suitable approach for customer churn prediction, where historical data with known labels (churn or non-churn) are available for training the model. Supervised learning enables the model to learn the relationship between input variables and the target outcome, allowing it to accurately predict whether a customer is likely to churn based on past behaviour.

Additionally, effective decision-making and strategy development depend heavily on the interpretability of customer churn prediction models. Interpretability refers to the extent to which model predictions can be understood by business users. Furthermore, the ability to evaluate customer churn prediction models is crucial for making wise decisions and developing strategies. Although sophisticated models like ensemble approaches and boosting algorithms often achieve accurate prediction performance, their complex nature limit transparency and makes it challenging to identify the primary reasons of churn (Boukrouh & Azmani, 2025; De Caigny et al., 2024). Therefore, this study compares Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and XGBoost to evaluate the balance between predictive performance and interpretability in e-commerce churn prediction.

Despite the growing application of machine learning techniques in customer churn prediction, several research gaps remain. Previous studies have mainly focused on improving predictive performance by developing models with higher accuracy, while less attention has been given to the interpretability of these models (Baghla & Gupta, 2022; Gordini & Veglio, 2017). Although advanced algorithms such as ensemble learning methods can effectively capture complex customer behaviour patterns, their complex structures often make it difficult for business stakeholders to understand the reasons behind churn predictions. Furthermore, limited studies have comprehensively compared multiple machine learning algorithms while integrating explainable artificial intelligence techniques to provide transparent insights into customer churn behaviour. Therefore, this study addresses the identified gap by evaluating multiple supervised machine learning models and applying SHapley Additive exPlanations (SHAP) based interpretability analysis to achieve both accurate prediction and meaningful explanations for e-commerce customer churn.

1.2 Problem Statement

Customer retention remains a significant challenge for many businesses, particularly those operating in the e-commerce sector. Although existing customers contribute greater long-term value, acquiring new customers requires substantial financial investment to build loyalty and sustain engagement (Alshamsi, 2022). A key issue for e-commerce companies is the difficulty in retaining profitable customers who may switch to competing platforms for various reasons. Therefore, it is essential for businesses to identify potential churners in advance and implement appropriate retention strategies, as failing to do so can lead to financial losses, wasted organisational resources, and long-term sustainability risks. Moreover, the lack of physical interaction between e-commerce companies and customers makes it more difficult to monitor customer satisfaction and engagement levels. Factors such as product quality, service performance, financial transactions, delivery efficiency, and delivery delays significantly influence customers' decisions to continue or discontinue using a platform, potentially resulting in the loss of valuable customers (Baghla & Gupta, 2022).

In the e-commerce industry, modelling customer retention is a complex and multidimensional task that involves various operational and technological challenges. One major issue is the heterogeneity of customer data, which originates from multiple sources such as transaction histories, browsing behaviour, and social media interactions. The large volume of data, inconsistent formats, and varying levels of data quality make it difficult to integrate and analyse these datasets effectively in order to extract meaningful patterns (Yaragani, 2020). In addition, customer behaviour on e-commerce platforms changes rapidly over time due to shifting consumer preferences, market competition, and promotional activities, making churn prediction more challenging. Furthermore, although advanced machine learning models are capable of achieving high prediction accuracy, many of these models lack interpretability and operate as "black box" systems, making it difficult for business managers to understand the factors influencing customer churn predictions. These challenges may reduce the effectiveness of predictive models and limit businesses' ability to implement timely, accurate, and data-driven customer retention strategies.

1.3 Research Objective

The main objective of this study is to develop, evaluate, and compare machine learning models for predicting customer churn in the e-commerce industry, with an emphasis on both predictive performance and model interpretability. Customer churn has become a significant challenge for e-commerce businesses as losing existing customers can negatively affect revenue generation, customer lifetime value, and long-term business sustainability. Therefore, this study aims to analyse customer behavioural and transactional data to identify important patterns and factors associated with customer churn behaviour.

To achieve this objective, this study develops customer churn prediction models using six supervised machine learning algorithms, namely Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and XGBoost. These models are evaluated and compared using various performance metrics, including accuracy, precision, recall, F1-score, confusion matrix, and Area Under the Receiver Operating

Characteristic Curve (AUC-ROC) to determine the most effective model for distinguishing between churn and non-churn customers.

Furthermore, this study applies SHapley Additive exPlanations (SHAP) analysis to improve the interpretability of machine learning predictions. While advanced machine learning models can achieve high predictive performance, their complex structures may limit understanding of the factors influencing prediction outcomes. Therefore, SHAP is used to identify the contribution of individual features towards churn predictions and provide meaningful insights into customer behaviour patterns.

Overall, this study aims to bridge the gap between predictive accuracy and practical application by developing machine learning models that can not only predict customer churn effectively but also provide transparent explanations to support data-driven decision-making in e-commerce customer retention management.

1.4 Significance of the Study

This study contributes to both theoretical and practical aspects of customer churn prediction in the e-commerce industry. From a theoretical perspective, this study extends existing research by conducting a comprehensive comparison of multiple machine learning models and integrating SHAP-based interpretability analysis. While previous studies often focus mainly on improving predictive performance, this study highlights the importance of balancing prediction accuracy with model transparency. The findings also contribute to a deeper understanding of how customer behavioural and transactional characteristics influence churn outcomes, providing additional insights into customer decision-making patterns within e-commerce environments.

From a practical perspective, this study provides valuable support for e-commerce businesses in developing a more data-driven approach to customer retention management. By evaluating different machine learning algorithms, this study helps organisations understand the strengths and suitability of various predictive approaches for identifying potential churn customers. Furthermore, the integration of SHAP improves the transparency of machine learning predictions by explaining the contribution of important features towards churn outcomes. This allows stakeholders to better understand prediction results and increases confidence in applying machine learning techniques for business decision-making.

2.0 Literature Review

2.1 Customer Churn in E-commerce Industry

In the e-commerce sector, customer churn is characterized by a transition where consumers cease purchasing goods or services from a company in favor of its competitors, often as a result of poor product quality or delayed deliveries. Since e-commerce functions under a non-contractual framework, unlike contractual sectors, it is essentially challenging for businesses to recognise or predict the quiet termination of a customer relationship before it happens (Alshamsi, 2022). Customer churn has become a major concern for e-commerce businesses because losing existing customers can negatively affect profitability and long-term business sustainability. Therefore, businesses are concentrating more on understanding customer behaviour and implementing effective retention strategies to reduce churn rates and maintain customer loyalty.

Several studies have highlighted that retaining existing customers is significantly more cost effective than acquiring new customers. This is due to the fact that while loyal consumers are more likely to make repeat purchases and contribute to consistent revenue development, businesses must invest more in advertising, promotions, and marketing efforts to attract new customers (Baghla & Gupta, 2022; Gordini & Veglio, 2017). In contrast, high customer churn rate will negatively impact e-commerce performance by reducing revenue, lowering Customer Lifetime Value (CLV), and increasing the costs of acquiring new customers, ultimately weakening overall business stability (Yilmaz, 2023). In addition, Xiahou & Harada (2022) highlight that even a small improvement in customer retention can generate substantial financial gains, as a 5 percent increase in retention may lead to a 25 percent to 95 percent increase in net present value and a 25 percent to 85 percent rise in profit margins, highlighting the financial impact of customer churn management.

Furthermore, customer churn in e-commerce is closely associated with the quality of customer relationships and the level of loyalty developed over time. Customers who receive good shopping experiences and satisfactory services are more likely to remain loyal and continue purchasing from the same e-commerce platform (Matuszelański & Kopczewska, 2022). In contrast, issues such as delayed deliveries, poor customer service and unresolved complaints can negatively affect customer satisfaction and trust, leading customers to switch to competing platforms that offer better experiences and services (Yaragani, 2020). Therefore, ensuring consistent service quality and maintaining strong customer trust are essential for strengthening customer loyalty and reducing the likelihood of customer churn in the e-commerce industry.

2.2 Overview of Customer Churn Prediction in E-commerce Industry

Customer churn prediction refers to the process of identifying customers who are likely to stop purchasing products or services from an e-commerce platform in the future by using historical customer data, behavioural patterns, and machine learning techniques. The main objective of customer churn prediction is to help businesses detect potential churners early so that appropriate retention strategies can be implemented to reduce customer loss and improve long-term profitability (Xiahou & Harada, 2022). In the e-commerce industry, customer churn prediction has become increasingly important because online customers can easily switch to competing platforms due to the non-contractual nature of e-commerce services. Therefore, businesses utilise predictive analytics and machine learning algorithms to analyse customer behaviours such as purchase frequency, transaction history, browsing activities, product reviews, and customer engagement in order to identify customers with a high risk of churn (Alshamsi, 2022).

Additionally, the development of customer churn prediction in the e-commerce industry has significantly transformed how companies understand and manage customer churn. E-commerce enterprises can easily obtain customer data directly from their databases and platform systems. This data typically includes demographic information such as age, gender, address, and contact details, as well as behavioural data such as shopping time, purchase frequency, product views, and cart additions (Xiahou & Harada, 2022). As a result, companies are now able to analyse customer behaviour more effectively and make data driven decisions to reduce churn. Researchers have emphasized that analyzing customer data enables organizations to predict customers who are likely to churn and implement proactive retention strategies before customers permanently leave the platform (De Caigny et al., 2024; Gordini & Veglio, 2017; Karimi et al., 2024).

Furthermore, previous studies have emphasized that customer churn prediction plays a significant role in supporting strategic decision making in e-commerce businesses. By accurately identifying customers who are likely to churn, organisations are able to design targeted retention strategies such as personalised promotions, loyalty programmes, and recommendation systems to enhance customer engagement and reduce customer attrition (Shobana et al., 2023). In addition, prior research has shown that churn prediction enables more effective marketing segmentation by identifying high-value customer groups based on their purchasing behaviour and interaction patterns. This allows businesses to tailor marketing activities according to customer preferences and improve campaign effectiveness. Researcher has also highlighted that timely and personalised marketing interventions, particularly during peak shopping periods, can significantly enhance customer response and encourage repeat purchases (Xiahou & Harada, 2022). As a result, customer churn prediction not only improves retention strategies but also supports efficient resource allocation and contributes to long-term business sustainability in the e-commerce sector.

2.3 Application of Machine Learning in Customer Churn Prediction

Machine learning plays an important role in customer churn prediction by enabling e-commerce companies to analyse large and complex datasets and identify customers who are likely to discontinue using a platform. Previous studies have shown that churn prediction is commonly framed as a classification problem, where customers are categorized into churn or non-churn groups based on their behavioural patterns (Baghla & Gupta, 2022; De Caigny et al., 2024; Yu et al., 2011). In this study, six machine learning models are employed, namely Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and XGBoost to evaluate and compare their performance in predicting customer churn.

Previous research found that Logistic Regression achieved approximately 90% prediction accuracy in identifying e-commerce customers who were likely to churn. The study highlighted that Logistic Regression was effective in predicting customer churn because the model could clearly classify customers into churn and non-churn groups based on customer behavioural data (Sunarya et al., 2024). Besides, a study conducted by Kim & Lee (2022) found that the Decision Tree model also achieved approximately 90% prediction accuracy in identifying churn customers in influencer commerce. The authors found that the Decision Tree model helped analyse customer purchase frequency and transaction behaviour, enabling businesses to identify customers with a higher possibility of discontinuing purchases and customers with stronger loyalty behaviour.

According to Sunarya et al. (2024), the Random Forest model achieved approximately 95% prediction accuracy in identifying customers who were likely to churn in e-commerce platforms. The authors found that the Random Forest model successfully identified customers with low application usage, low transaction frequency, and lower satisfaction levels as customers with a higher possibility of churn. In addition, previous research showed that the SVM model achieved higher prediction accuracy than Logistic Regression in identifying churn customers in B2C e-commerce platforms. The authors found that the SVM model produced an average prediction accuracy of 91.56% after customer segmentation and showed better prediction performance for churn classification compared to the Logistic Regression model (Xiahou & Harada, 2022).

Furthermore, Awasthi (2022) found that the KNN model achieved high recall and AUC-ROC scores in predicting e-commerce customer churn, indicating that the model was effective in correctly identifying customers who were likely to leave the platform. The study also highlighted that KNN performed better than SVM and Random Forest in terms of recall performance, demonstrating its capability in analysing customer behavioural similarities and improving churn classification accuracy. Furthermore, previous studies have shown that XGBoost is highly effective in customer churn prediction because it can analyse complex and non-linear customer behaviour patterns more accurately than traditional machine learning models. Hermawan et al. (2025) found that the XGBoost model achieved strong prediction performance with a weighted average F1-score of 0.91 and an AUC value of 0.9583, demonstrating its capability in accurately identifying customers with high churn risk and supporting proactive customer retention strategies.

2.4 Handling Imbalanced Data in Customer Churn Prediction

Imbalanced data is a common problem in customer churn prediction studies, particularly in e-commerce datasets where the number of non-churn customers is significantly larger than the number of churn customers. In real-world e-commerce environments, only a relatively small proportion of customers discontinue using a platform, resulting in unequal class distribution within the dataset. Previous research found that imbalanced data negatively affects the performance of customer churn prediction models because machine learning algorithms tend to focus more on the majority non-churn class while ignoring the minority churn class (Suguna et al., 2025). Consequently, predictive models may produce misleadingly high accuracy while demonstrating poor performance in identifying actual churn customers, reducing the reliability of churn prediction and customer retention strategies.

To overcome data imbalance problems, Synthetic Minority Oversampling Technique (SMOTE) is one of the most widely applied methods in customer churn prediction studies. SMOTE generates synthetic samples of the minority class (churned customers) to create a more balanced dataset, preventing the models from being biased toward the majority class and improving predictive accuracy (Mubarak et al., 2023).

According to previous studies in the e-commerce industry, the application of SMOTE can improve class balance and enhance the predictive performance of machine learning models by reducing overfitting issues commonly associated with random oversampling methods (Alshamsi, 2022; Yilmaz, 2023). Suguna et al. (2025) further demonstrated that SMOTE effectively balances the minority churn class by generating synthetic customer data, enabling machine learning models to better identify churn cases. Their findings showed that SMOTE improved balanced accuracy, weighted recall, and F1-score across multiple machine learning models, resulting in better churn prediction performance on imbalanced datasets.

2.5 Model Interpretability in Customer Churn Prediction

Model interpretability has become an increasingly important consideration in customer churn prediction research because businesses require not only accurate predictions but also clear explanations behind those predictions. Traditional statistical approaches generally provide higher transparency because the relationship between input variables and outcomes can be directly examined (Bhattacharjee et al., 2026). However, the increasing adoption of advanced machine learning algorithms, particularly ensemble and boosting techniques has improved the ability to capture complex patterns from large-scale customer datasets (Imani et al., 2025; Manzoor et al., 2024). Despite their strong predictive capabilities, these models often contain complex internal structures that make it difficult for users to understand how individual features influence the final prediction. This limitation is commonly referred to as the “black-box” problem, where a model can produce highly accurate results but provides limited explanation regarding the reasoning behind its decisions (Peng et al., 2023).

In the e-commerce industry, the lack of transparency in machine learning models may reduce the practical value of churn prediction outcomes. Identifying customers with a high probability of churn alone is insufficient if businesses are unable to understand the factors contributing to such predictions. Understanding whether churn is associated with declining engagement, reduced purchase frequency, changes in customer behaviour, or other factors allows businesses to develop more appropriate retention strategies. Therefore, model interpretability plays an important role in converting machine learning predictions into meaningful business insights (Manzoor et al., 2024). By providing explanations behind prediction outcomes, interpretable models can improve stakeholder confidence, support better decision-making, and encourage the adoption of machine learning techniques in customer relationship management.

Although machine learning techniques have been widely applied in customer churn prediction, existing studies still reveal limitations regarding the integration of predictive performance and model interpretability (Ekanayake & De Alwis, 2025). Many previous studies have primarily focused on improving classification accuracy and comparing model performance, while less emphasis has been placed on explaining the factors that drive churn predictions (Alshamsi, 2022; Baghla & Gupta, 2022; Gordini & Veglio, 2017). Although advanced algorithms such as ensemble learning models can effectively identify complex customer behaviour patterns, their lack of transparency may limit their acceptance and application among business stakeholders (Haider et al., 2025). Therefore, a research gap exists in developing churn prediction approaches that can achieve high predictive performance while also providing understandable explanations of customer churn behaviour. Addressing this gap is essential because businesses require not only reliable predictions but also actionable insights to support effective customer retention decisions.

To address these limitations, SHapley Additive exPlanations (SHAP) is widely adopted as an interpretability technique in customer churn prediction. SHAP is a model agnostic explainability method based on cooperative game theory that assigns each feature a contribution value to explain its influence on model predictions. This enables both global and local interpretability, which allows businesses to understand overall model behaviour as well as the specific reasons behind individual churn predictions, thereby improving transparency and supporting more informed and trustworthy decision making (Ardhani & Tania, 2025).

Besides, SHAP assists researchers in identifying the most influential features contributing to customer churn, enabling companies to design targeted retention strategies. By improving model transparency and interpretability, SHAP enhances stakeholder confidence in machine learning predictions and supports the implementation of predictive analytics in e-commerce environments (Boukrouh & Azmani, 2025). De Caigny et al. (2024) stated that SHAP helps improve the interpretability of black-box customer churn prediction models by explaining how each feature contributes to churn predictions at both global and local levels. Their findings showed that SHAP provided insights into customer churn behaviour, enabling businesses to better understand churn patterns and support more effective customer retention strategies.

3.0 Methodology

3.1 Data Collection

The dataset used in this study is the E commerce Customer Churn Dataset obtained from Kaggle. It consists of 6,000 customer records and includes 15 input variables together with one target variable indicating customer churn. The dataset is suitable for classification tasks as it is designed for predictive modelling of customer churn in an e commerce environment. Although the dataset is synthetic, it is carefully constructed to reflect realistic customer behaviour patterns commonly observed in real world e-commerce platforms. The dataset is structured to simulate typical customer lifecycle information and is widely used for machine learning research purposes. This structured format makes the dataset appropriate for developing and evaluating multiple machine learning classification models.

3.2 Data Understanding

The dataset consists of 15 input variables (independent variables) and 1 target variable (dependent variable). These variables capture various aspects of customer behavior and interaction with the e-commerce platform. The input variables include customer ID, account age months, average order value, total orders, days since last purchase, discount usage rate, return rate, customer support tickets, loyalty member, browsing frequency per week, cart abandonment rate, product review score average, engagement score, satisfaction score and price sensitivity index whereas the target variable is churned. Table 1 shows the detailed information about the dataset.

Table 1: Details of attributes

Attributes	Data Type	Description
Customer ID	Numerical	Unique identifier for each customer
Account age months	Numerical	Duration of customer relationship with the platform
Average order value	Numerical	Average monetary value of customer purchases
Total orders	Numerical	Total number of orders placed by the customer
Days since last purchase	Numerical	Number of days since the last purchase
Discount usage rate	Numerical	Percentage of using discounts or promotions

Return rate	Numerical	Percentage of returned items
Customer support tickets	Numerical	Number of support requests raised
Loyalty member	Categorical	Indicates whether the customer is enrolled in a loyalty program
Browsing frequency per week	Numerical	Frequency of platform visits
Cart abandonment rate	Numerical	Percentage of shopping carts abandoned without completing the purchase
Product review score average	Numerical	Average rating given by the customer
Engagement score	Numerical	Overall engagement level with the platform
Satisfaction score	Numerical	Customer satisfaction level
Price sensitivity index	Numerical	Sensitivity to price changes
Churned	Categorical	Indicates whether a customer has discontinued using the platform

3.3 Data Preparation

Data preparation is an essential step in machine learning studies because the quality of the dataset directly influences the performance and reliability of predictive models. Proper data preparation helps improve data quality, ensure model compatibility, and enhance predictive performance by handling issues such as missing values, outliers, imbalanced data, and inconsistent feature representations (Mladenovic et al., 2026).

The first step involved conducting Exploratory Data Analysis (EDA), a preliminary data analysis technique used to understand the structure, distribution, patterns, and characteristics of a dataset prior to further analysis and model development (Dhummad, 2025). In this study, EDA was performed to gain an initial understanding of the dataset before any modelling decisions were made. The dataset structure was examined through shape inspection and data type verification, while descriptive statistics were generated to summarise the central tendency, spread, and range of each variable.

The second step involved data preprocessing, which is regarded as one of the most critical stages in any data analysis pipeline, as it focuses on transforming raw data containing inconsistencies, missing values, noise, and outliers into a clean and structured format suitable for analysis (Çetin & Yıldız, 2022). In this study, any identifier columns that offered no predictive value were removed to ensure only relevant features were retained, and missing values were checked across all features to ensure data completeness. The independent variable 'loyalty member' and the target variable 'churned' were then label-encoded into binary form, where 'Yes' is represented as 1 and 'No' is represented as 0.

The third step involved splitting the dataset into training and testing sets, using an 80:20 ratio with stratification on the target variable to preserve the original class proportions in both subsets (Joseph, 2022). This step was deliberately carried out prior to outlier treatment, feature selection, and feature scaling, as performing these procedures on the full dataset before splitting would allow information from the testing set to influence the training process, a methodological issue known as data leakage. By splitting the data first, all subsequent transformations could be fitted exclusively on the training set, ensuring that the testing set remained completely unseen and provided an unbiased evaluation of model performance.

The fourth step involved outlier detection and treatment. Boxplot visualisation was first applied to the training data to identify the distribution of numerical variables and provide an early indication of extreme values. The Interquartile Range (IQR) method was then used to compute the lower and upper bounds from the training set only, and these bounds were applied to cap extreme values in both the training and testing sets (Abuzaid & Alkronz, 2024). This approach was adopted to preserve the overall data distribution while minimising the influence of extreme values, thereby improving data quality and supporting more reliable model training, without allowing the testing set to influence the capping thresholds.

The fifth step involved feature selection, which is a process in machine learning that identifies the most relevant variables from a large set of input features to improve model performance, reduce overfitting, and enhance computational efficiency (Hamdard & Lodin, 2023). In this study, feature selection was conducted using a Random Forest classifier trained exclusively on the training set, which is an ensemble learning method that evaluates the contribution of each feature by measuring the average reduction in impurity across all decision trees (Menze et al., 2009). Feature importance scores were extracted and used to rank all input variables, and features that fell below an importance threshold of 0.01 were considered to have negligible predictive value and were subsequently removed. The same set of selected features was then applied to the testing set, ensuring consistency between the two subsets.

The sixth step involved feature scaling, which is a preprocessing technique used to standardise the range of independent variables so that all features contribute equally to the model and improve learning performance (Pinheiro et al., 2025). In this study, StandardScaler was fitted on the training set only to learn the mean and standard deviation of each feature, and the resulting transformation was then applied to both the training and testing sets to standardise all numerical features to a mean of 0 and a standard deviation of 1, while preventing information from the testing set from influencing the scaling parameters.

The seventh step involved addressing class imbalance in the target variable using the Synthetic Minority Over-sampling Technique (SMOTE). This technique was applied exclusively to the training set to generate synthetic samples of the minority class, thereby creating a more balanced distribution for model training, while the testing set was left untouched to preserve its original class distribution and ensure a realistic evaluation of model performance.

3.4 Modeling

This study applies six models, namely Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and XGBoost to predict customer churn. These models are selected to allow comparison in terms of prediction performance and interpretability.

3.4.1 Logistic Regression

Logistic Regression known as the logistic or logit model, is a statistical and machine learning classification technique used to examine the relationship between multiple independent variables and a categorical dependent variable. It estimates the probability of a specific outcome by fitting the data to a logistic curve. Unlike Linear Regression, logistic regression does not predict continuous values. Instead, it calculates the log odds of an outcome and transforms it into a probability using the logistic sigmoid function, ensuring that the predicted values fall within the range of 0 to 1 (Sperandei, 2014).

3.4.2 Decision Tree

Decision Tree is a tree structure that resembles a flowchart, with each internal node representing an attribute test, each branch representing the test's result, and each leaf node (or terminal node) representing the class label (Sharma & Kumar, 2016). It is widely used in machine learning and data mining due to its simplicity and interpretability. The model works by applying a series of tests, where features are compared to threshold values to guide classification. Compared to complex models like neural networks, decision trees are easier to understand and construct. Decision Tree is mainly used for classification tasks, grouping data based on feature values, and consist of nodes and branches that represent features and their possible outcomes (Charbuty & Abdulazeez, 2021).

3.4.3 Random Forest

Random Forest is a supervised machine learning method that works for issues involving both regression and classification. Regression problems are solved by the average prediction, whereas classification decisions are made by the majority of votes (Vergni & Todisco, 2023). It is built on an ensemble of multiple decision trees, which generally provides higher accuracy than a single decision tree model. To introduce diversity among the trees, random forest applies bootstrap aggregation and random feature selection during training. This approach makes the model robust and effective for handling both categorical and continuous data (Sarker, 2021).

3.4.4 Support Vector Machine (SVM)

Support Vector Machine (SVM), introduced by Vapnik in 1979, is a supervised machine learning algorithm initially developed for binary classification and later extended to regression tasks known as Support Vector Regression (SVR). It works by identifying an optimal hyperplane that separates data points with the maximum margin, making it effective for classification problems in complex datasets (Rodríguez-Pérez & Bajorath, 2022). SVM is widely recognised for its strong generalisation ability, particularly with limited training data. When data are not linearly separable, kernel functions such as radial basis function and polynomial kernels are used to transform the input space into higher dimensions to improve class separation. However, although SVM can achieve high predictive performance, its computational complexity increases with larger datasets, making it more resource intensive in terms of processing and memory requirements (Abdullah & Abdulazeez, 2021).

3.4.5 K-Nearest Neighbors (KNN)

The K-Nearest Neighbor (KNN) algorithm is a simple and widely used method for classification and regression in machine learning and data mining, with applications in areas such as pattern recognition, text categorisation, object detection, and event recognition. It is a non-parametric algorithm that does not assume any specific data distribution, making it suitable for problems where the underlying structure is unknown. KNN is often used as a baseline model due to its simplicity and effectiveness, particularly when limited prior knowledge of the dataset is available. It is also considered a lazy learning algorithm because it stores the training data and performs computation only during prediction (Abu Alfeilat et al., 2019). The classification process is based on identifying the K nearest data points in the feature space using a distance metric and assigning the class label through majority voting (Feng et al., 2023).

3.4.6 XGBoost

XGBoost stands for Extreme Gradient Boosting. The XGBoost algorithm is an optimized version of the Gradient Boosting Decision Tree (GBDT) method developed by Chen and Guestrin (2016), designed to improve the efficiency and predictive performance of traditional GBDT models. It works by constructing an objective function and optimizing it iteratively to achieve the best predictive model. This objective function consists of two main parts: a loss function and a regularization term. The loss function measures the difference between predicted values and actual outcomes, while the regularization term controls model complexity and helps reduce the risk of overfitting (Liang et al., 2025).

3.5 Evaluation Metrics

This study evaluates machine learning models using key performance metrics, including accuracy, precision, recall, and F1-score. These metrics are selected because they are commonly used in classification tasks and are particularly important for imbalanced datasets such as customer churn prediction. Each metric provides a different perspective on model performance, allowing for a more comprehensive evaluation than accuracy alone. Therefore, using multiple evaluation measures ensures a more reliable and meaningful comparison of the models (Sujon et al., 2025).

3.5.1 Confusion matrix

A confusion matrix is a table used to assess the performance of a classification model by comparing actual labels with predicted labels. It is organized in rows for actual classes and columns for predicted classes, allowing a clear view of correct classifications and errors. This helps in understanding how well a model performs across different categories and where misclassifications occur (Yang & Berdine, 2024). Figure 1 presents the confusion matrix used to evaluate the classification model.

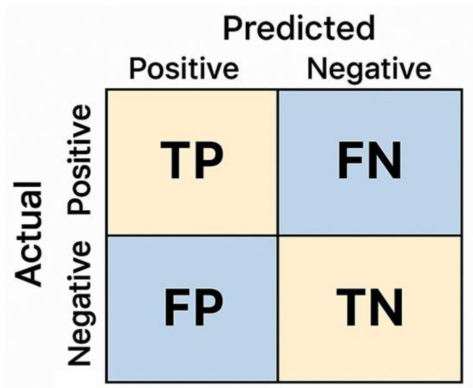

Figure 1: Confusion matrix

There are four types of outcomes in a confusion matrix:

i. **True positive (TP)** occur when the model correctly predicts a positive class.
ii. **True negative (TN)** occur when the model correctly predicts a negative class.
iii. **False positive (FP)** occur when the model incorrectly predicts a positive class while the actual class is negative.
iv. **False negative (FN)** occur when the model incorrectly predicts a negative class while the actual class is positive.

3.5.2 Accuracy

Accuracy is a commonly used evaluation metric in classification models that indicates how often a model makes correct predictions. It measures the proportion of correctly classified cases out of the total number of predictions. In other words, it compares the number of true positives and true negatives against all observations to assess overall model performance (Sathyanarayanan & Tantri, 2024). The formula for accuracy is presented below:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}}$$

3.5.3 Recall

Recall evaluates how well a classification model identifies actual positive cases. It is computed as the number of true positives divided by the sum of true positives and false negatives. This metric indicates the model's ability to correctly capture all relevant positive instances in the dataset (Schlosser et al., 2024). The formula for recall is shown below:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

3.5.4 Precision

Precision used to assess how accurate a model's positive predictions. It is determined by dividing the number of true positives by the total number of predicted positives, which includes both true positives and false positives. This measure reflects how often the model's positive predictions are actually correct, indicating the reliability of its classification results (Schlosser et al., 2024). The formula for precision is shown below:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

3.5.5 F1-score

The F1-score is a performance metric that merges precision and recall into one overall measure using their harmonic mean. It is designed to evaluate a model's performance in a balanced way, particularly in cases where both false positive and false negative errors need to be considered (Sathyanarayanan & Tantri, 2024). The formula for F1-score is shown below:

$$\text{F1} - \text{score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}$$

3.5.6 Area Under the Receiver Operating Characteristic Curve (AUC-ROC)

Area Under the Receiver Operating Characteristic Curve (AUC-ROC) is used to evaluate the performance of a classification model. The ROC curve plots the true positive rate against the false positive rate at different threshold levels, while the AUC measures the overall ability of the model to distinguish between classes, where a higher value indicates better performance (Schlosser et al., 2024). The formula for AUC-ROC is shown below:

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR})\, d(\text{FPR})$$

4.0 Empirical Results

4.1 Result from pre-processing steps

4.1.1 Data Cleaning and Label Encoding

As shown in Figure 2, data cleaning is conducted to prepare the dataset for analysis. The 'Customer ID' column is removed because it does not provide useful information for prediction. The dataset is then checked for missing values to ensure there are no incomplete records.

Categorical variables are subsequently transformed into numerical format using label encoding. The 'loyalty member' variable is encoded where ‘Yes’ is represented as 1 and ‘No’ is represented as 0. The target variable 'churned' is also encoded in the same way, where ‘Yes’ is 1 and ‘No’ is 0. This step ensures that only the most relevant variables are formatted correctly and used to predict customer churn.

```
Customer_ID column removed successfully.

Missing values in each column:
account_age_months            0
avg_order_value               0
total_orders                  0
days_since_last_purchase      0
discount_usage_rate           0
return_rate                   0
customer_support_tickets      0
loyalty_member                0
browsing_frequency_per_week   0
cart_abandonment_rate         0
product_review_score_avg      0
engagement_score              0
satisfaction_score            0
price_sensitivity_index       0
churned                       0
dtype: int64

Result: No missing values were found in the dataset.
```

Figure 2: Missing Value Check Results

4.1.2 Data Splitting

As shown in Figure 3, the dataset was split into an 80% training set (4,800 rows) and a 20% testing set (1,200 rows) using the ‘train_test_split function’ from the Scikit-learn library, with both sets retaining 14 independent features. A random state of 42 ensured reproducibility, while stratification was used to maintain the original balance of churned and non-churned customers in both sets. These sets are used to train and validate the machine learning models for customer churn prediction.

```
X_train shape: (4800, 14)
X_test shape : (1200, 14)
```

Figure 3: Training and Testing Dataset Splitting Results

4.1.3 Outlier Detection and Capping

Outlier detection was performed using the Interquartile Range (IQR) method alongside boxplot visualisations, with the operational boundaries computed strictly from the training set to prevent data leakage. The binary variable 'loyalty member' was excluded from this process, as it is a categorical 0/1 indicator rather than a continuous variable, making IQR-based outlier detection inapplicable. As shown in Figure 4, initial visualisations identified clear outlier distributions across multiple variables. Based on the statistical summary in Figure 5, the highest proportion of flagged values was observed in 'customer support tickets' at 6.81%, followed by 'days since last purchase' at 5.00%, and 'average order value' at 4.67%. Features such as 'account age months' and 'cart abandonment rate' contained zero outliers from the outset.

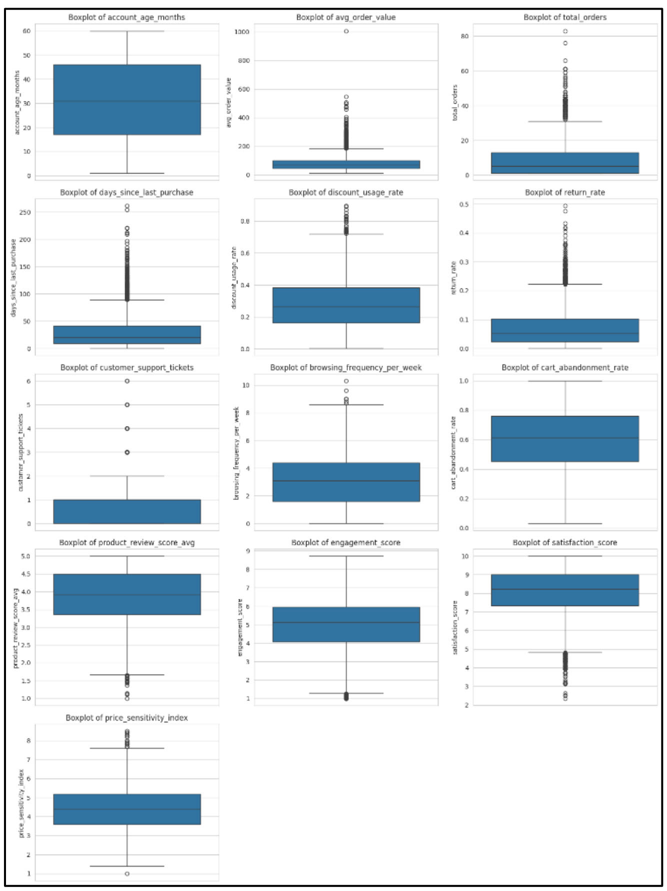

Figure 4: Boxplots of numerical features before capping

Outlier counts before capping:

Column	Lower Bound	Upper Bound	Outlier Count (before capping)	Outlier % (before capping)
customer_support_tickets	-1.50	2.50	327	6.81
days_since_last_purchase	-39.00	89.00	240	5.00
avg_order_value	-39.26	184.35	224	4.67
return_rate	-0.10	0.22	172	3.58
total_orders	-17.00	31.00	159	3.31
engagement_score	1.27	8.76	131	2.73
satisfaction_score	4.81	11.53	73	1.52
discount_usage_rate	-0.17	0.72	36	0.75
price_sensitivity_index	1.20	7.60	35	0.73
product_review_score_avg	1.66	6.18	21	0.44
browsing_frequency_per_week	-2.60	8.60	8	0.17
account_age_months	-26.50	89.50	0	0.00
cart_abandonment_rate	-0.01	1.22	0	0.00

Figure 5: Outlier counts before IQR capping

To prevent these extreme values from distorting model training, IQR-based capping was applied to both the training and testing sets using the frozen bounds derived exclusively from the training data. As detailed in Figure 6, this processing successfully reduced the outlier count across all features to 0.0%. This capping approach successfully controlled the influence of extreme data points while preserving the original dataset size and maintaining a clear separation between the training parameters and the unseen testing data.

Outlier counts after capping:

Column	Lower Bound	Upper Bound	Outlier Count (after capping)	Outlier % (after capping)
account_age_months	-26.50	89.50	0	0.0
avg_order_value	-39.26	184.35	0	0.0
total_orders	-17.00	31.00	0	0.0
days_since_last_purchase	-39.00	89.00	0	0.0
discount_usage_rate	-0.17	0.72	0	0.0
return_rate	-0.10	0.22	0	0.0
customer_support_tickets	-1.50	2.50	0	0.0
browsing_frequency_per_week	-2.60	8.60	0	0.0
cart_abandonment_rate	-0.01	1.22	0	0.0
product_review_score_avg	1.66	6.18	0	0.0
engagement_score	1.27	8.76	0	0.0
satisfaction_score	4.81	11.53	0	0.0
price_sensitivity_index	1.20	7.60	0	0.0

Figure 6: Outlier counts after IQR capping

4.1.4 Feature Selection

Feature selection was performed using a Random Forest classifier to identify the most relevant variables for predicting customer churn. The classifier evaluated the contribution of each feature by measuring how much it reduced impurity across the decision trees. A threshold of 0.01 was applied to determine feature relevance, where variables with importance values below this threshold were considered less meaningful and removed from the dataset. The results indicated that most features were strong predictors of churn, reflecting their overall relevance in explaining customer behaviour. Figure 7 presents the feature importance ranking for all variables used in the customer churn prediction model, highlighting the relative contribution of each feature to the prediction performance. The ‘loyalty member’ feature was the only variable excluded, as its importance score of 0.0081 fell below the 0.01 threshold. This score indicates that the variable provided minimal predictive value or influence when distinguishing between customers who churned and those who stayed. The final selected feature set therefore retained the majority of variables, ensuring that key behavioural and transactional information was preserved for model training while maintaining a more efficient and focused dataset.

```
Feature Importance Ranking (from training data):
engagement_score               0.4713
days_since_last_purchase       0.3161
satisfaction_score             0.0360
browsing_frequency_per_week    0.0257
product_review_score_avg       0.0244
cart_abandonment_rate          0.0184
return_rate                    0.0173
discount_usage_rate            0.0167
avg_order_value                0.0147
account_age_months             0.0146
price_sensitivity_index        0.0137
customer_support_tickets       0.0118
total_orders                   0.0112
loyalty_member                 0.0081
dtype: float64
```

Figure 7: Feature importance ranking

4.1.5 Feature Scaling

Feature scaling was performed using the StandardScaler technique to normalize all selected features model training. This method standardizes each feature by transforming it to have a mean equal to 0 and a standard deviation of 1, thereby minimizing the effect of differences in feature magnitudes. To prevent data leakage, the scaler was fitted strictly to the training data predictor variables, and the transformation was then applied to both datasets. To maintain interpretability, the scaled data were converted back into a DataFrame while preserving the original column names. As shown in Figure 8, the descriptive statistics of the standardized training features indicate that all mean values are at 0.0 and standard deviations are at 1.0, confirming that the scaling process was successfully completed. This preprocessing step ensures that the machine learning models are trained on features with comparable scales, thereby improving model stability and performance.

	count	mean	std	min	25%	50%	75%	max
engagement_score	4800.0	-0.0	1.0	-2.47	-0.56	0.16	0.72	2.61
days_since_last_purchase	4800.0	-0.0	1.0	-1.13	-0.77	-0.33	0.51	2.44
satisfaction_score	4800.0	0.0	1.0	-2.73	-0.64	0.12	0.76	1.58
browsing_frequency_per_week	4800.0	0.0	1.0	-1.63	-0.78	0.01	0.70	2.92
product_review_score_avg	4800.0	0.0	1.0	-2.89	-0.68	0.04	0.78	1.45
cart_abandonment_rate	4800.0	-0.0	1.0	-2.82	-0.73	0.07	0.79	1.97
return_rate	4800.0	-0.0	1.0	-1.17	-0.79	-0.28	0.56	2.58
discount_usage_rate	4800.0	-0.0	1.0	-1.80	-0.78	-0.12	0.65	2.79
avg_order_value	4800.0	-0.0	1.0	-1.55	-0.75	-0.22	0.53	2.46
account_age_months	4800.0	-0.0	1.0	-1.73	-0.81	-0.00	0.86	1.67
price_sensitivity_index	4800.0	-0.0	1.0	-2.90	-0.79	-0.09	0.61	2.72
customer_support_tickets	4800.0	-0.0	1.0	-0.95	-0.95	0.25	0.25	2.05
total_orders	4800.0	0.0	1.0	-0.84	-0.84	-0.38	0.54	2.60

Figure 8: Descriptive statistics of features after standard scaling

4.1.6 Class Balancing using SMOTE

To address the class imbalance within the dataset, the Synthetic Minority Oversampling Technique (SMOTE) was applied strictly to the training data after feature scaling. This approach effectively balanced the target classes while preventing any potential data leakage into the testing set. As illustrated in Figure 9, before applying SMOTE, the training dataset contained 4,057 non-churned customers and 743 churned customers. After oversampling, both classes were balanced equally with 4,057 instances each, resulting in a total of 8,114 training records. This balancing process enhanced the learning capability of the machine learning models by reducing bias toward the majority class and improving prediction performance for churned customers.

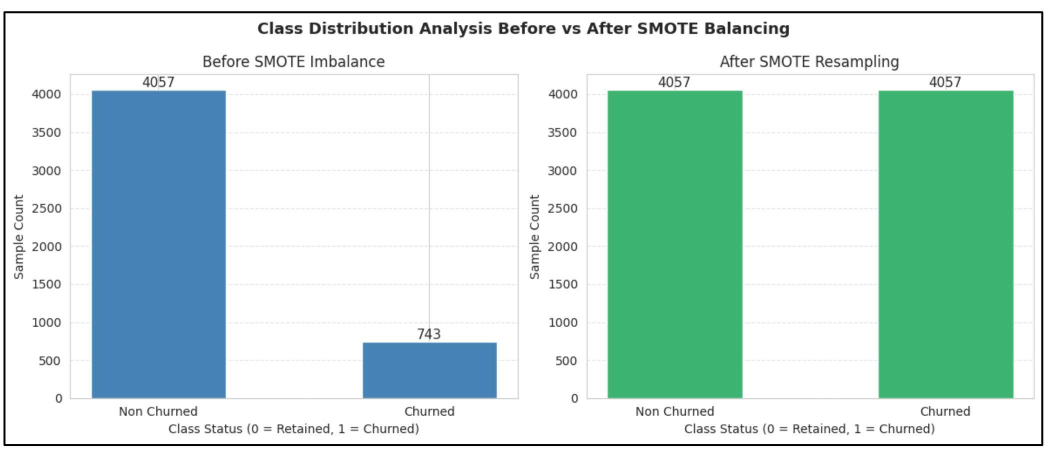

Figure 9: Class Distribution Before and After Applying SMOTE

4.2 Comparative performance of the machine learning models

4.2.1 Comparison of Confusion Matrix

Table 2 presents the confusion matrix results for the six machine learning models, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and XGBoost. The confusion matrix evaluates model performance by comparing the actual customer classes with the predicted classes.

Logistic Regression correctly classified 970 non-churn customers and 177 churn customers, while producing 44 false positive and 9 false negative predictions. These results indicate that the model achieved a strong ability to identify churn customers while maintaining relatively low classification errors. The Decision Tree model correctly predicted 974 non-churn customers and 163 churn customers; however, it generated a higher number of false negatives (23 cases) compared to Logistic Regression, suggesting a greater possibility of overlooking customers who were likely to churn.

Random Forest demonstrated consistent classification performance by correctly identifying 982 non-churn customers and 172 churn customers, with only 32 false positives and 14 false negatives. This indicates that the ensemble-based approach was effective in reducing prediction errors and improving customer classification reliability. Similarly, the SVM model achieved 972 correct predictions for non-churn customers and 173 correct predictions for churn customers, although it recorded 42 false positive cases.

The K-Nearest Neighbors (KNN) model successfully classified 947 non-churn customers and 177 churn customers. However, it produced the highest number of false positives (67 cases), indicating a weaker ability to distinguish non-churn customers compared with the other models. In contrast, XGBoost achieved the strongest overall classification performance, correctly identifying 988 non-churn customers and 166 churn customers while maintaining the lowest false positive rate among the evaluated models. Although XGBoost recorded a slightly higher number of false negatives compared with some models, its overall classification capability demonstrated strong effectiveness in separating churn and non-churn customers.

Overall, the confusion matrix results provide an initial comparison of the classification behaviour and error patterns across the six machine learning models. These multi-model matrix breakdowns serve as the underlying foundation for the subsequent evaluation of accuracy, precision, recall, and F1-score, which provide more comprehensive performance benchmarks for determining the most suitable model for customer churn prediction.

Table 2: Confusion matrix of six machine learning models

Models	Actual	Predicted	
		No	Yes
Logistic Regression	No	970	44
	Yes	9	177
Decision Tree	No	974	40
	Yes	23	163
Random Forest	No	982	32
	Yes	14	172
Support Vector Machine (SVM)	No	972	42
	Yes	13	173
K-Nearest Neighbors (KNN)	No	947	67
	Yes	9	177
XGBoost	No	988	26
	Yes	20	166

4.2.2 Comparison of Accuracy

Table 3 presents the accuracy results of the six machine learning models. **XGBoost** and **Random Forest** achieved the highest accuracy of **96.2%**, indicating the strongest overall classification performance among the evaluated models. Logistic Regression followed with an accuracy of 95.6%, while SVM and Decision Tree achieved 95.4% and 94.8% respectively. KNN recorded the lowest accuracy of 93.7%, suggesting comparatively weaker classification capability due to its distance-based prediction approach. Overall, the results show that ensemble-based models, particularly XGBoost and Random Forest provided the most accurate predictions among the evaluated approaches.

Table 3: Accuracy of six machine learning models

Models	Accuracy
Logistic Regression	0.956
Decision Tree	0.948
Random Forest	**0.962**
Support Vector Machine (SVM)	0.954
K-Nearest Neighbors (KNN)	0.937
XGBoost	**0.962**

4.2.3 Comparison of Precision, Recall, F1-score, and AUC Score

Table 4 presents the performance evaluation of the six machine learning models using precision, recall, F1-score, and AUC score. These metrics provide a more detailed comparison of how effectively each model identifies churn customers and distinguishes them from non-churn customers. In terms of precision, XGBoost achieved the highest score of 0.865, followed by Random Forest with 0.843. This indicates that XGBoost was more accurate when predicting customers as churners, as a higher precision value means fewer non-churn customers were incorrectly classified as churn. On the other hand, KNN recorded the lowest precision score of 0.725, showing that although it could identify churn customers, it produced more false positive predictions compared with the other models.

For recall, Logistic Regression and KNN achieved the highest score of 0.952, indicating that these models were able to capture the largest proportion of actual churn customers. This is important in churn prediction because failing to identify potential churners may result in missed opportunities for customer retention. Random Forest also demonstrated strong recall performance with a value of 0.925, while XGBoost recorded a lower recall of 0.892, suggesting that it missed slightly more actual churn cases compared with other models.

The F1-score evaluates the balance between precision and recall, making it an important metric when both correct churn identification and reducing incorrect predictions are required. Random Forest achieved the highest F1-score of 0.882, followed by XGBoost with 0.878 and

Logistic Regression with 0.870. This shows that Random Forest provided the most balanced performance in identifying churn customers while controlling false predictions. Meanwhile, KNN achieved the lowest F1-score of 0.823 due to its lower precision performance.

Figure 10 presents the Receiver Operating Characteristic (ROC) curves and corresponding Area Under the Curve (AUC) scores for the six machine learning models. Models with curves closer to the top left corner indicate better classification performance and stronger ability to distinguish between churn and non-churn customers. Logistic Regression achieved the highest AUC score of 0.993, followed closely by Random Forest and XGBoost, both with an AUC score of 0.992, while SVM obtained an AUC score of 0.991. These models exhibited ROC curves that approached the top left corner, indicating excellent discriminative capability. KNN achieved an AUC score of 0.970, whereas Decision Tree recorded the lowest AUC score of 0.918, suggesting comparatively weaker classification performance. Overall, the ROC curve analysis demonstrates that Logistic Regression, Random Forest, XGBoost, and SVM provided excellent discrimination between churn and non-churn customers, with Logistic Regression exhibiting the strongest overall performance.

Based on the overall evaluation, **Random Forest** was selected as the champion model for customer churn prediction. Although XGBoost achieved the highest precision, Random Forest provided a better overall balance by achieving the highest F1-score (0.882), strong recall (0.925), and the joint-highest AUC score (0.992). Since customer churn prediction requires identifying as many potential churn customers as possible while maintaining reliable predictions, Random Forest was considered the most suitable model for supporting effective customer retention strategies.

Table 4: Precision, Recall, F1-score, and AUC Score of six machine learning models

Models	**Precision**	**Recall**	**F1-Score**	**AUC Score**
Logistic Regression	0.801	0.952	0.870	0.993
Decision Tree	0.803	0.876	0.838	0.918
Random Forest	**0.843**	**0.925**	**0.882**	**0.992**
Support Vector Machine (SVM)	0.805	0.930	0.863	0.991
K-Nearest Neighbors (KNN)	0.725	0.952	0.823	0.970
XGBoost	0.865	0.892	0.878	0.992

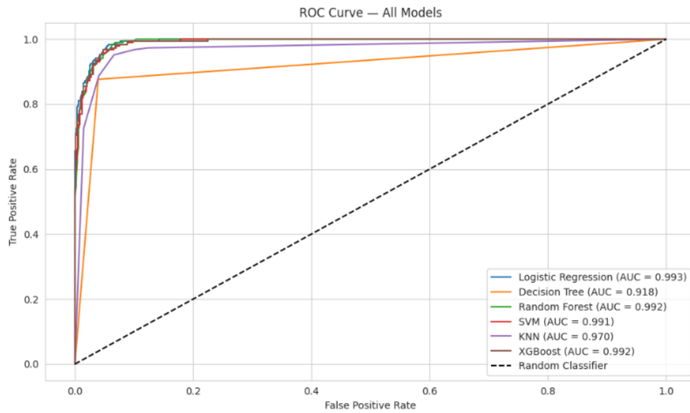

Figure 10: ROC Curves for six machine learning models

4.2.4 Result Analysis using SHAP

Figure 11 presents the SHapley Additive exPlanations (SHAP) Feature Importance Bar Plot based on the selected Random Forest champion model. SHAP was employed to quantify the average contribution of each feature to the model predictions and enhance model interpretability. The results indicate that **engagement score** is the most influential feature affecting customer churn prediction, followed by **days since last purchase**. Other variables such as satisfaction score, customer support tickets, browsing frequency per week, and product review score average also contributed to the prediction process, but with relatively lower importance compared to the most influential features. In contrast, variables such as price sensitivity index and discount usage rate showed relatively smaller impacts on the model predictions. Overall, the SHAP analysis provides greater interpretability by identifying the most significant factors influencing customer churn behaviour and explaining how individual features contribute to the prediction outcomes.

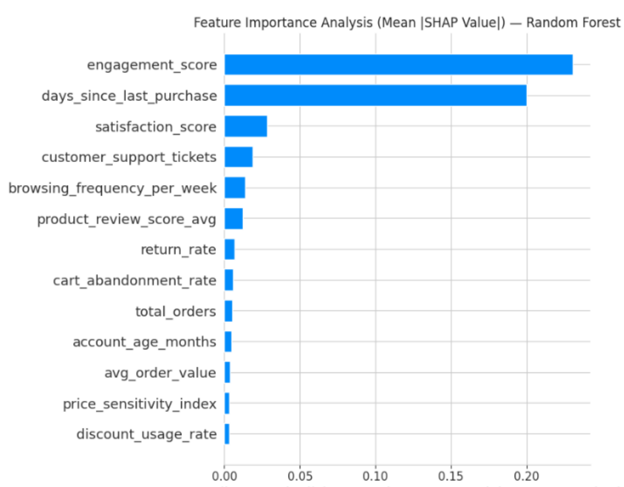

Figure 11: SHAP Feature Importance Bar Plot

Figure 12 presents the SHAP Summary Plot, which ranks features by their contribution to churn prediction while revealing the direction and magnitude of each variable's impact. The analysis identifies **engagement score** as the most influential predictor, where low engagement (blue) drives positive SHAP values that increase churn probability, while high engagement (red) exerts a strong protective effect. **Days since last purchase** emerges as the second most critical driver, with longer purchase intervals (red) consistently pushing predictions toward churn, confirming that customer recency is a powerful behavioral signal. Additionally, **satisfaction score** and **customer support tickets** demonstrate notable effects, where low satisfaction and high-ticket volumes both correlate positively with increased churn risk, while satisfied customers and those with fewer support interactions show negative SHAP values indicating retention. Overall, the SHAP summary plot enhances the interpretability of the machine learning model by providing clear insights into how different customer behavior factors contribute to churn prediction in the e-commerce platform.

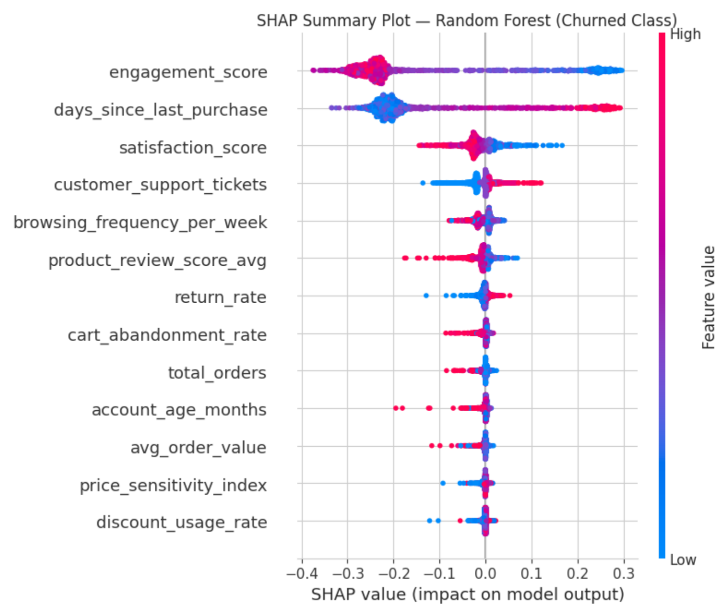

Figure 12: SHAP Summary Plot

5.0 Discussion

The findings of this study demonstrate that machine learning techniques are highly effective in predicting customer churn within the e-commerce environment. Based on the experimental results, **XGBoost** and **Random Forest** both achieved the highest accuracy of 96.2%. The strong performance of these models indicates their capability to effectively identify patterns associated with customer churn behaviour. In particular, the superior performance of XGBoost suggests that advanced ensemble learning methods are highly suitable for churn prediction tasks, as they can capture complex and nonlinear relationships within customer behavioural data while maintaining strong generalisation performance. Besides, the strong performance of Random Forest indicates its robustness in handling high-dimensional customer behavioural data by aggregating multiple decision trees, which improves model stability and reduces sensitivity to individual data variations. These findings are consistent with previous studies which reported that ensemble learning methods such as XGBoost and Random Forest are highly reliable for customer churn prediction in e commerce environments due to their ability to combine multiple decision trees, which enhances prediction stability, reduces overfitting, and improves overall classification performance in complex customer behavioural datasets (Kumar, 2026; Sun, 2025).

Expanding upon these accuracy findings, a deeper examination of the classification metrics including precision, recall, F1-scores, and Receiver Operating Characteristic (ROC) curves and Area Under the Curve (AUC) scores further solidifies **Random Forest** as the absolute champion model among all evaluated algorithms. While high overall accuracy can sometimes be misleading in classification tasks involving imbalanced data, the Random Forest model demonstrated exceptional balance across every evaluated metric. It achieved a precision of 0.843, a strong recall of 0.925, and the highest F1-score of 0.882 for the minority churn class while maintaining low misclassification errors in the confusion matrix. Furthermore, the ROC curves illustrated that Random Forest achieved an exceptionally high AUC score of 0.992, indicating superior threshold-independent discriminative ability. These findings strongly align with prior literature, with Random Forest highlighting the operational power of ensemble classifiers over individual baseline algorithms in e-commerce analytics. Specifically, the outstanding predictive metrics of Random Forest echo the work of Fadhila et al. (2024), who applied Recency, Frequency, and Monetary (RFM) framework to fashion e-commerce and found that Random Forest outperformed all other classifiers with an optimal accuracy of 0.99 and an AUC score of 0.98. Similarly, a comparative evaluation by Foree et al. (2022) using digital retail records confirmed that Random Forest established the highest classification accuracy (85%) and AUC score (0.927) among competing pipelines. Furthermore, the model's ability to maximize classification performance while keeping false-positive and false-negative errors to a minimum directly supports the conclusions of Ardhani & Tania, 2025, who stated that tree-based ensemble methods are optimized for capturing customer attrition patterns in complex e-commerce environments. According to these authors, traditional individual estimators often experience high prediction variance or generalize poorly when encountering dense, overlapping behavioral data spaces. By contrast, tree-based ensembles such as Random Forest resolve these limitations because they are able to capture non-linear relationships within customer behavioral records while reducing the risk of overfitting through the aggregation of multiple decision boundaries. Therefore, the empirical evidence from this study supports the academic consensus that Random Forest and other ensemble methods provide a stable and effective framework for tracking digital customer relationships over time.

Based on the SHapley Additive exPlanations (SHAP) Feature Importance Bar Plot (Figure 11) and SHAP Summary Plot (Figure 12), the empirical results reveal that **engagement score** acts as the primary driving feature influencing customer churn predictions, followed closely by **days since last purchase**. The directional distribution within the summary plot explicitly demonstrates that lower platform **engagement scores** (represented by blue data points) significantly accelerate the positive log-odds of a customer churning. Logically, this indicates that a lower engagement score suggests customers are becoming less active on the platform, which is a strong sign that they are losing interest and gradually distancing themselves from the business. This primary behavioral outcome strongly reinforces the findings of Liu and Liu (2026), who concluded that user behavioral metrics related to digital engagement serve as the most vital predictor when capturing online shoppers' retention habits. From a consumer behavior perspective, this drop in engagement represents the initial stage of psychological detachment. In a highly competitive e-commerce landscape where consumers face minimal switching costs, a decline in active platform interactions, such as fewer product clicks or reduced session durations, signifies that the platform has lost its cognitive share of the consumer's attention and is likely yielding to competitor alternatives.

Concurrently, SHAP Summary Plot (Figure 12) demonstrates that high values of **days since last purchase** (represented by red data points) strongly push predictions toward the churn outcome. This indicates that longer periods without making a purchase are a strong signal that customers are becoming inactive and more likely to leave the platform. In other words, the more time that passes since a customer’s last transaction, the higher the risk of churn. This dynamic closely matches the empirical evidence from Hang (2026), which argued that a higher number of days since a customer's last purchase functions as a primary indicator of customer disengagement and platform abandonment. Together, these two features map out a clear, sequential journey of customer attrition. The dwindling engagement score provides an early-stage warning of fading interest, while the growing number of days since the last purchase acts as the actual behavioral proof of churn. This dynamic suggests that when a customer breaks their transactional habit with a platform for too long, standard algorithmic marketing triggers like automated push notifications and generic discount emails lose their power, making re-engagement highly challenging. Ultimately, the application of SHAP supports the interpretability of the Random Forest model by reducing the “black-box” nature of machine learning and clearly explaining how customer behavioural features influence churn predictions. Rather than simply flagging a customer as a high churn risk, this level of transparency empowers marketing teams to move beyond generalized campaigns and instead deploy highly targeted retention interventions that address the exact reasons behind a customer's disengagement.

From a theoretical perspective, this study directly advances business analytics and customer relationship management literature by resolving the historic conflict between model accuracy and transparency. While traditional research often assumes that elite predictive power requires sacrificing clarity, this paper proves that pairing an advanced Random Forest architecture with game-theoretic SHAP frameworks eliminates the need for a “black-box” compromise. It establishes a clean, scalable academic blueprint for how complex data science workflows can maintain a high threshold independent discriminative ability while providing deep structural interpretability. Furthermore, it contributes to consumer behavior theory by empirically validating that digital behavioral micro indicator, specifically platform engagement metrics and transactional time gaps, serve as superior, dynamic leading signals of customer attrition compared to static demographic data.

Practically, the findings of this study provide valuable insights for e-commerce companies to develop more effective customer retention strategies. The SHAP analysis identified

engagement score and **days since last purchase** as the two most influential factors affecting customer churn prediction, indicating that declining customer interaction and longer periods of inactivity are strong warning signals of potential churn. Based on these findings, companies can implement early detection mechanisms by continuously monitoring customer engagement levels and purchase recency to identify high-risk customers before they leave the platform. Customers with decreasing engagement scores can be targeted through personalised retention campaigns, such as tailored product recommendations, loyalty rewards, exclusive promotions, or re-engagement messages to encourage continued interaction. Similarly, customers with longer periods since their last purchase can be approached through targeted win-back strategies, including personalised discounts, customer feedback surveys, or service improvements to understand and address the reasons behind their inactivity. By focusing retention efforts on these key behavioural indicators, businesses can move from reactive customer recovery to proactive churn prevention, allowing them to allocate marketing resources more efficiently and improve long-term customer lifetime value.

Despite the valuable findings obtained from this study, several limitations should be acknowledged. First, this research utilised a synthetic e-commerce customer churn dataset obtained from Kaggle, which may not fully capture the complexity and variability of real-world customer behaviours. Although the dataset contains realistic behavioural and transactional attributes, actual e-commerce environments are influenced by various external factors such as market changes, economic conditions, competitor activities, and seasonal purchasing patterns, which were not considered in this study. Second, the analysis was limited to structured customer behavioural and transactional variables, while other important information such as demographic characteristics, social media interactions, and real-time browsing activities were excluded. The absence of these additional data sources may reduce the ability of the models to capture a more comprehensive understanding of customer churn behaviour. Third, this study focused on supervised machine learning models and SHAP-based interpretation, without exploring advanced approaches such as deep learning, time-series modelling, or real-time prediction systems that may better adapt to changing customer behaviours. Therefore, future research should validate the proposed framework using real-world multi-source datasets and investigate more advanced predictive techniques to improve model generalisation and practical application in dynamic e-commerce environments.

6.0 Conclusion and Recommendations

This study successfully developed and evaluated six supervised machine learning models, namely Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and XGBoost to predict customer churn in the e-commerce industry. The findings demonstrate that machine learning techniques are highly effective in identifying customers who are likely to discontinue using an e-commerce platform based on behavioural and transactional data. Among all evaluated models, **Random Forest** emerged as the **champion model** by achieving the highest overall performance. It reached an overall classification accuracy of **96.2%**, a highly competitive precision of **0.843**, a dominant recall of **0.925**, a peak F1-score of **0.882**, and an outstanding Area Under the Curve (AUC) score of **0.992**. The strong predictive performance of the Random Forest model can be attributed to its ensemble learning approach based on bootstrap aggregating (bagging). By constructing multiple decision trees using different subsets of the training data and combining their outputs through majority voting, the model is able to reduce prediction variance and minimize the risk of overfitting commonly associated with a single decision tree. This enables the model to capture complex and non-linear relationships among customer behavioural variables more effectively. Overall, the Random Forest model developed in this study offers a robust and reliable approach for identifying customers at risk of churning, allowing businesses to take timely retention actions before permanent revenue loss occurs.

In addition, the application of SHapley Additive exPlanations (SHAP) significantly improved the interpretability of the prediction model by identifying the most influential variables contributing to customer churn. The results revealed that **engagement score** and **days since last purchase** were the most important factors influencing churn behaviour. Specifically, customers who showed lower levels of activity on the platform and had not made purchases for a longer period were more likely to discontinue using the e-commerce service. These findings provide meaningful business insights that can help e-commerce companies design targeted retention strategies such as personalised marketing campaigns, loyalty programmes, and customer engagement initiatives. Furthermore, SHAP improves model transparency by clearly explaining how each feature contributes to the final prediction, allowing managers and non-technical stakeholders to understand why certain customers are classified as high churn risk. This reduces the “black-box” nature commonly associated with machine learning models and increases trust in data-driven decision-making. Overall, this study contributes to both academic research and practical business analytics by demonstrating that combining predictive performance with model interpretability can support more transparent, accurate, and data-driven decision-making in customer relationship management.

Based on the limitations identified in this study, several recommendations are proposed for future research. First, future studies should utilise real-world customer datasets collected from actual e-commerce platforms instead of relying only on synthetic datasets. Real-world data obtained from different time periods and business environments would provide a more accurate representation of customer behaviour by capturing changes in purchasing patterns, market conditions, and seasonal trends. Furthermore, applying the proposed churn prediction framework across different e-commerce platforms or industries could improve the generalisability and reliability of the findings.

Second, future research should incorporate additional customer-related information from multiple data sources to provide a more comprehensive understanding of churn behaviour. While this study focused mainly on behavioural and transactional variables, future studies may include demographic information, customer feedback, social media interactions, website browsing behaviour, and real-time engagement activities. These additional features may help identify more complex customer preferences and improve the ability of machine learning models to detect early signs of churn.

Third, future studies are encouraged to investigate more advanced predictive approaches beyond traditional supervised machine learning models. Techniques such as deep learning, time-series modelling, hybrid machine learning frameworks, and real-time prediction systems may provide stronger capabilities in capturing evolving customer behaviours over time. In addition, future research may explore automated machine learning and continuous model updating approaches to improve model adaptability in dynamic e-commerce environments. Combining advanced predictive techniques with explainable artificial intelligence methods such as SHAP or other interpretation approaches can further enhance both prediction performance and business trust in machine learning-based customer retention decisions.

7.0 References

Abdullah, D. M., & Abdulazeez, A. M. (2021). Machine learning applications based on SVM classification a review. *Qubahan Academic Journal*, *1*(2), 81-90. https://doi.org/10.48161/qaj.v1n2a50

Abu Alfeilat, H. A., Hassanat, A. B., Lasassmeh, O., Tarawneh, A. S., Alhasanat, M. B., Eyal Salman, H. S., & Prasath, V. S. (2019). Effects of distance measure choice on k-nearest neighbor classifier performance: a review. *Big data*, *7*(4), 221-248. https://doi.org/10.1089/big.2018.0175

Abuzaid, A., & Alkronz, E. (2024). A comparative study on univariate outlier winsorization methods in data science context. *Statistica Applicata-Italian Journal of Applied Statistics*, (1). https://doi.org/10.26398/IJAS.0036-004

Alnuaimi, A. F., & Albaldawi, T. H. (2024). An overview of machine learning classification techniques. In *BIO Web of Conferences* (Vol. 97, p. 00133). EDP Sciences. https://doi.org/10.1051/bioconf/20249700133

Alshamsi, A. (2022). Customer Churn prediction in ECommerce Sector. https://repository.rit.edu/theses/11183

Awasthi, S. (2022). Customer churn prediction on e-commerce data using stacking classifier. *Authorea Preprints*. https://doi.org/10.36227/techrxiv.20291694.v1

Baghla, S., & Gupta, G. (2022). Performance evaluation of various classification techniques for customer churn prediction in e-commerce. *Microprocessors and Microsystems*, *94*, 104680. https://doi.org/10.1016/j.micpro.2022.104680

Bhattacharjee, B., Madhu, U., Guha, S. K., Bhadra, S., Das, P. K., Samantaray, S. P., ... & Tamboli, S. (2026). Neural network approach enhancing churn prediction with categorical encoding and standard scaling. *Scientific Reports*. https://doi.org/10.1038/s41598-026-37407-z

Boukrouh, I., & Azmani, A. (2025). Explainable machine learning models applied to predicting customer churn for e-commerce. *Int J Artif Intell ISSN*, *2252*(8938), 8938. https://doi.org/10.11591/ijai.v14.i1.pp286-297

Çetin, V., & Yıldız, O. (2022). A comprehensive review on data preprocessing techniques in data analysis. *Pamukkale Üniversitesi Mühendislik Bilimleri Dergisi*, *28*(2), 299-312. http://doi.org/10.5505/pajes.2021.62687

Charbuty, B., & Abdulazeez, A. (2021). Classification based on decision tree algorithm for machine learning. *Journal of applied science and technology trends*, *2*(01), 20-28. https://doi.org/10.38094/jastt20165

Chen, T., & Guestrin, C. (2016, August). Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining* (pp. 785-794). https://doi.org/10.1145/2939672.2939785

De Caigny, A., De Bock, K. W., & Verboven, S. (2024). Hybrid black-box classification for customer churn prediction with segmented interpretability analysis. *Decision Support Systems*, *181*, 114217. https://doi.org/10.1016/j.dss.2024.114217

Dhummad, S. (2025). The imperative of exploratory data analysis in machine learning. *Scholars Journal of Engineering and Technology*, *13*(01), 30-44. https://doi.org/10.36347/sjet.2025.v13i01.005

Ekanayake, I., & De Alwis, S. (2025). Explainability, risk modeling, and segmentation based customer churn analytics for personalized retention in e-commerce. *arXiv preprint arXiv:2510.11604*. https://doi.org/10.48550/arXiv.2510.11604

Fadhila, H. L., Permadi, V. A., & Tahalea, S. P. (2024). Optimising the Fashion E-Commerce Journey: A Data-Driven Approach to Customer Retention. *Knowledge Engineering and Data Science*, *7*(1), 5. https://doi.org/10.17977/um018v7i12024p58-70

Feng, C., Zhao, B., Zhou, X., Ding, X., & Shan, Z. (2023). An enhanced quantum K-nearest neighbor classification algorithm based on polar distance. *Entropy*, *25*(1), 127. https://doi.org/10.3390/e25010127

Foree, N. D., Westerbeek, R., de Andrade Serra, P. J., & Hoogendoorn, M. (2022). Machine learning predictions of customer churn from store and e-commerce data.

Gordini, N., & Veglio, V. (2017). Customers churn prediction and marketing retention strategies. An application of support vector machines based on the AUC parameter-selection technique in B2B e-commerce industry. *Industrial Marketing Management*, *62*, 100-107. https://doi.org/10.1016/j.indmarman.2016.08.003

Haider, R., Bari, M. F. A. I., Afia, N., & Karim, T. (2025). Illuminating the black box: Explainable AI for enhanced customer behavior prediction and trust. *International Journal of Science and Research Archive*, *15*(3), 247-268. https://doi.org/10.30574/ijsra.2025.15.3.1674

Hamdard, M. S., & Lodin, H. (2023). Effect of feature selection on the accuracy of machine learning model. *International Journal of Multidisciplinary Research and Analysis*, *6*(9), 4460-4466. https://doi.org/10.47191/ijmra/v6-i9-66

Hang, J. (2026). Customer churn prediction model for E-commerce using an enhanced adaptive boosting algorithm. *Discover Applied Sciences*. https://doi.org/10.1007/s42452-026-08745-4

Hermawan, A., Saputra, A., Rafi, M. D., Basmallah, S., Putra, Y. T. S., & Nabila, W. (2025). Implementing xgboost model for predicting customer churn in e-commerce platforms. *Repeater: Publikasi Teknik Informatika dan Jaringan*, *3*(2), 17-31. https://doi.org/10.62951/repeater.v3i2.401

Imani, M., Joudaki, M., Beikmohammadi, A., & Arabnia, H. R. (2025). Customer churn prediction: A systematic review of recent advances, trends, and challenges in machine learning and deep learning. *Machine Learning and Knowledge Extraction*, *7*(3), 105. https://doi.org/10.3390/make7030105

Jain, V., Malviya, B. I. N. D. O. O., & Arya, S. A. T. Y. E. N. D. R. A. (2021). An overview of electronic commerce (e-Commerce). Journal of Contemporary Issues in Business and Government, 27(3), 666. https://doi.org/10.47750/cibg.2021.27.03.090

Joseph, V. R. (2022). Optimal ratio for data splitting. *Statistical Analysis and Data Mining: The ASA Data Science Journal*, *15*(4), 531-538. https://doi.org/10.1002/sam.11583

Karimi, M. U., Abubakar, S. M., Mustafa, S. J., & Ahmad, B. (2024). Introduction to Artificial Intelligence and Machine Learning Algorithms: A Review. *International Journal of Advanced Natural Sciences and Engineering Researches*, *8*(5), 30–34. Retrieved from https://as-proceeding.com/index.php/ijanser/article/view/1881

Kedah, Z. (2023). Use of e-commerce in the world of business. *Startupreneur Business Digital (SABDA Journal)*, *2*(1), 51-60. https://doi.org/10.33050/sabda.v2i1.273

Kim, S., & Lee, H. (2022). Customer churn prediction in influencer commerce: An application of decision trees. *Procedia Computer Science*, *199*, 1332-1339. https://doi.org/10.1016/j.procs.2022.01.169

Kumar, P. S. (2026). Predicting Customer Churn in E-Commerce Using Machine Learning: A Comprehensive Approach. *International Journal of Informatics and Information Systems*, *9*(2), 357-370. https://doi.org/10.47738/ijiis.v9i2.300

Liang, B., Qin, W., & Liao, Z. (2025). A Differential Evolutionary-Based XGBoost for Solving Classification of Physical Fitness Test Data of College Students. *Mathematics*, *13*(9), 1405. https://doi.org/10.3390/math13091405

Liu, J., & Liu, C. (2026). Prediction of Online Shoppers' Purchasing Intention Based on SHAP-IGWO-EM Method. *International Journal of Swarm Intelligence Research (IJSIR)*, *17*(1), 1-28. https://doi.org/10.4018/IJSIR.397667

Manzoor, A., Qureshi, M. A., Kidney, E., & Longo, L. (2024). A review on machine learning methods for customer churn prediction and recommendations for business practitioners. *IEEE access*, *12*, 70434-70463. https://doi.org/10.1109/ACCESS.2024.3402092

Matuszelański, K., & Kopczewska, K. (2022). Customer churn in retail e-commerce business: Spatial and machine learning approach. *Journal of Theoretical and Applied Electronic Commerce Research*, *17*(1), 165-198. https://doi.org/10.3390/jtaer17010009

Menze, B. H., Kelm, B. M., Masuch, R., Himmelreich, U., Bachert, P., Petrich, W., & Hamprecht, F. A. (2009). A comparison of random forest and its Gini importance with standard chemometric methods for the feature selection and classification of spectral data. *BMC bioinformatics*, *10*(1), 213. https://doi.org/10.1186/1471-2105-10-213

Mladenovic, S., Lindauer, M., & Doerr, C. (2026). Automated Data Preparation for Machine Learning: A Survey. *Journal of Data-centric Machine Learning Research*.

Mubarak, M. M. R., Chrisnanto, Y. H., & Sabrina, P. N. (2023). Implementation of Random Forest Using Smote and Smoteenn in Customer Churn Classification in E-Commerce. *Enrichment: Journal of Multidisciplinary Research and Development*, *1*(8), 463-477. https://doi.org/10.55324/enrichment.v1i8.69

Peng, K., Peng, Y., & Li, W. (2023). Research on customer churn prediction and model interpretability analysis. *Plos one*, *18*(12), e0289724. https://doi.org/10.1371/journal.pone.0289724

Pinheiro, J. M. H., de Oliveira, S. V. B., Silva, T. H. S., Saraiva, P. A. R., de Souza, E. F., Godoy, R. V., ... & Becker, M. (2025). The impact of feature scaling in machine learning: Effects on regression and classification tasks. *IEEE Access*, *13*, 199903-199931. https://doi.org/10.1109/ACCESS.2025.3635541

Rodríguez-Pérez, R., & Bajorath, J. (2022). Evolution of support vector machine and regression modeling in chemoinformatics and drug discovery. *Journal of Computer-Aided Molecular Design*, *36*(5), 355-362. https://doi.org/10.1007/s10822-022-00442-9

Sarker, I. H. (2021). Machine learning: Algorithms, real-world applications and research directions. *SN computer science*, *2*(3), 1-21. https://doi.org/10.1007/S42979-021-00592-X

Sathyanarayanan, S., & Tantri, B. R. (2024). Confusion matrix-based performance evaluation metrics. *African Journal of Biomedical Research*, *27*(4S), 4023-4031. https://doi.org/10.53555/AJBR.v27i4S.4345

Schlosser, T., Friedrich, M., Meyer, T., & Kowerko, D. (2024). A consolidated overview of evaluation and performance metrics for machine learning and computer vision. *Tobias Schlosser, Michael Friedrich, Trixy Meyer, and Danny Kowerko–Junior Professorship of*

Media Computing, Chemnitz University of Technology, *9107*. https://doi.org/10.13140/RG.2.2.14331.69928

Sharma, H., & Kumar, S. (2016). A survey on decision tree algorithms of classification in data mining. *International journal of science and research (IJSR)*, *5*(4), 2094-2097. https://doi.org/10.21275/v5i4.NOV162954

Shobana, J., Gangadhar, C., Arora, R. K., Renjith, P. N., Bamini, J., & devidas Chincholkar, Y. (2023). E-commerce customer churn prevention using machine learning-based business intelligence strategy. *Measurement: Sensors*, *27*, 100728. https://doi.org/10.1016/j.measen.2023.100728

Sperandei, S. (2014). Understanding logistic regression analysis. *Biochemia medica*, *24*(1), 12-18. https://doi.org/10.11613/BM.2014.003

Suguna, R., Suriya Prakash, J., Aditya Pai, H., Mahesh, T. R., Vinoth Kumar, V., & Yimer, T. E. (2025). Mitigating class imbalance in churn prediction with ensemble methods and SMOTE. *Scientific Reports*, *15*(1), 16256. https://doi.org/10.1038/s41598-025-01031-0

Sujon, K. M., Hassan, R., Choi, K., & Samad, M. A. (2025). Accuracy, precision, recall, f1-score, or MCC? empirical evidence from advanced statistics, ML, and XAI for evaluating business predictive models. *Journal of Big Data*, *12*(1), 268. https://doi.org/10.1186/s40537-025-01313-4

Sun, R. (2025, December). Customer Churn Prediction in E-Commerce Using Machine Learning Models for Digital Marketing Trends. In *Proceedings of the 2025 6th International Conference on Big Data Economy and Information Management* (pp. 994-999). https://doi.org/10.1145/3800000.3800153

Sunarya, P. A., Rahardja, U., Chen, S. C., Lic, Y. M., & Hardini, M. (2024). Deciphering digital social dynamics: A comparative study of logistic regression and random forest in predicting e-commerce customer behavior. *Journal of Applied Data Sciences*, *5*(1), 100-113. https://doi.org/10.47738/jads.v5i1.155

Tang, H. Y., & Ya'acob, S. (2023). E-Commerce Customer Churn Prediction for the Marketplace in Malaysia. *Open International Journal of Informatics*, *11*(2), 58-66. https://doi.org/10.11113/oiji2023.11n2.273

Vergni, L., & Todisco, F. (2023). A random forest machine learning approach for the identification and quantification of erosive events. *Water*, *15*(12), 2225. https://doi.org/10.3390/w15122225

Xiahou, X., & Harada, Y. (2022). B2C E-commerce customer churn prediction based on K-means and SVM. *Journal of Theoretical and Applied Electronic Commerce Research*, *17*(2), 458-475. https://doi.org/10.3390/jtaer17020024

Yang, S., & Berdine, G. (2024). Confusion matrix. *The Southwest Respiratory and Critical Care Chronicles*, *12*(53), 75-79. https://doi.org/10.12746/swrccc.v12i53.1391

Yaragani, V. K. (2020). Decoding E-commerce Customer Churn: Harnessing Data Science to Combat Negative Experiences. *North American Journal of Engineering Research*, *1*(4). https://najer.org/najer/article/view/10

Yilmaz, B. (2023). Application of Customer Churn Prediction Models in E-commerce. *BSc Thesis*.

Yu, X., Guo, S., Guo, J., & Huang, X. (2011). An extended support vector machine forecasting framework for customer churn in e-commerce. *Expert Systems with Applications*, *38*(3), 1425-1430. https://doi.org/10.1016/j.eswa.2010.07.049

Appendices

Part A: Python Source Code

```python
# Load data analysis libraries
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.ensemble import RandomForestClassifier

# Set style
sns.set_style('whitegrid')
```

Figure A1: Code for Importing Libraries and Setting Style

```python
# Load datasets
features_df = pd.read_csv('/content/drive/MyDrive/kaggle/input/ecommerce_customer_features.csv')
targets_df = pd.read_csv('/content/drive/MyDrive/kaggle/input/ecommerce_customer_targets.csv')

# Merge them on Customer_ID
df = pd.merge(features_df, targets_df, on='Customer_ID')

print(f"Dataset Shape: {df.shape}")
df.head()
```

Figure A2: Code for Loading and Merging Datasets

```python
# drop Customer_ID as it is not a predictive feature
df_clean = df.drop(['Customer_ID'], axis=1)
print("Customer_ID column removed successfully.")

# Check for missing values
missing_values = df_clean.isnull().sum()
print("\nMissing values in each column:")
print(missing_values)

# Display overall result
if missing_values.sum() == 0:
    print("\nResult: No missing values were found in the dataset.")
else:
    print(f"\nResult: A total of {missing_values.sum()} missing values were detected in the dataset.")
```

Figure A3: Code for Setting Up Stratified Data Splitting (80:20)

```python
# encode loyalty_member: Yes = 1, No = 0
df_clean['loyalty_member'] = df_clean['loyalty_member'].map({'Yes': 1, 'No': 0})

# encode churned (target): Yes = 1, No = 0
df_clean['churned'] = df_clean['churned'].map({'Yes': 1, 'No': 0})

df_clean.head(15)
```

Figure A4: Code for Label Encoding Categorical Variables

```python
from sklearn.model_selection import train_test_split

# Separate features (X) from the target label (y)
X = df_clean.drop('churned', axis=1)
y = df_clean['churned']

# Split into 80% training and 20% test; stratify to preserve class ratio
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

print(f'X_train shape: {X_train.shape}')
print(f'X_test shape : {X_test.shape}')
```

Figure A5: Code for Stratified Data Splitting (80:20)

```python
# Only look at numeric columns; binary columns don't have meaningful "outliers"
numeric_cols = X_train.select_dtypes(include='number').columns.tolist()
numeric_cols = [c for c in numeric_cols if c != 'loyalty_member']

# Plot individual boxplots for each numeric feature to visually inspect spread and outliers
plt.figure(figsize=(15, 20))
for i, col in enumerate(numeric_cols):
    plt.subplot(len(numeric_cols)//3 + 1, 3, i+1)
    sns.boxplot(y=X_train[col])
    plt.title(f'Boxplot of {col}')
plt.tight_layout()
plt.show()
```

Figure A6: Code for Generating Multi-Subplot Outlier Boxplots

```python
# Calculate IQR-based outlier bounds using TRAINING data only (no test-set leakage)
outlier_bounds  = {}
outlier_summary = []

for col in numeric_cols:
    Q1  = X_train[col].quantile(0.25)
    Q3  = X_train[col].quantile(0.75)
    IQR = Q3 - Q1

    # Standard Tukey fences: anything beyond 1.5x IQR from Q1/Q3 is flagged
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    outlier_bounds[col] = (lower, upper)

    # Count how many rows fall outside the bounds before any capping
    outlier_count = X_train[(X_train[col] < lower) | (X_train[col] > upper)].shape[0]
    outlier_pct   = (outlier_count / len(X_train)) * 100

    outlier_summary.append({
        'Column': col,
        'Lower Bound': round(lower, 2),
        'Upper Bound': round(upper, 2),
        'Outlier Count (before capping)': outlier_count,
        'Outlier % (before capping)': round(outlier_pct, 2)
    })

summary_df = pd.DataFrame(outlier_summary).sort_values('Outlier Count (before capping)', ascending=False)
print('Outlier counts before capping:')
print(summary_df.to_string(index=False))
```

Figure A7: Code for Calculating Frozen IQR Outlier Bounds

```python
# Apply IQR capping (Winsorization) using bounds learned from training data only
# The same bounds are applied to the test set to prevent data leakage
for col, (lower, upper) in outlier_bounds.items():
    X_train[col] = X_train[col].clip(lower, upper)
    X_test[col]  = X_test[col].clip(lower, upper)

# Re-check the training set to confirm capping worked (counts should now all be 0)
verify_summary = []
for col in numeric_cols:
    lower, upper  = outlier_bounds[col]
    outlier_count = X_train[(X_train[col] < lower) | (X_train[col] > upper)].shape[0]
    outlier_pct   = (outlier_count / len(X_train)) * 100

    verify_summary.append({
        'Column': col,
        'Lower Bound': round(lower, 2),
        'Upper Bound': round(upper, 2),
        'Outlier Count (after capping)': outlier_count,
        'Outlier % (after capping)': round(outlier_pct, 2)
    })

verify_df = pd.DataFrame(verify_summary)
print('Outlier counts after capping:')
print(verify_df.to_string(index=False))
```

Figure A8: Code for Applying Winsorization Outlier Capping

```python
# Train a Random Forest purely to rank feature importance — not used as the final model
rf_selector = RandomForestClassifier(random_state=42, n_estimators=100)
rf_selector.fit(X_train, y_train)

# Extract and sort each feature's importance score (higher = more predictive)
feat_importance = pd.Series(rf_selector.feature_importances_, index=X_train.columns)
feat_importance = feat_importance.sort_values(ascending=False)

print('Feature Importance Ranking (from training data):')
print(feat_importance.round(4))
```

Figure A9: Code for Baseline Random Forest Feature Selection

```python
# Drop features with importance below 0.01 — they contribute very little signal
low_importance     = feat_importance[feat_importance < 0.01].index.tolist()
selected_features = feat_importance[feat_importance >= 0.01].index.tolist()

print(f'Low importance features to drop: {low_importance}')
print(f'Selected features: {selected_features}')

# Apply the same feature subset to both train and test sets
X_train = X_train[selected_features]
X_test  = X_test[selected_features]

print(f'X_train shape after feature selection: {X_train.shape}')
print(f'X_test shape after feature selection : {X_test.shape}')
```

Figure A10: Code for Feature Dropping and Feature Matrix Slicing

```
from sklearn.preprocessing import StandardScaler

# Initialise StandardScaler – transforms each feature to zero mean and unit variance
scaler = StandardScaler()

# Fit the scaler on training data only, then apply the same transform to test data
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled  = scaler.transform(X_test)

# Convert arrays back to DataFrames to retain column names and index
X_train_scaled = pd.DataFrame(X_train_scaled, columns=X_train.columns, index=X_train.index)
X_test_scaled  = pd.DataFrame(X_test_scaled,  columns=X_test.columns,  index=X_test.index)

# Confirm scaling: mean ≈ 0 and std ≈ 1 for all features
X_train_scaled.describe().round(2).T
```

Figure A11: Code for Fitting and Transforming with StandardScaler

```
from imblearn.over_sampling import SMOTE

smote = SMOTE(random_state=42)
X_train_smote, y_train_smote = smote.fit_resample(X_train_scaled, y_train)

print(f'Before SMOTE: {X_train_scaled.shape}, class counts:\n{y_train.value_counts()}')
print(f'\nAfter SMOTE : {X_train_smote.shape}, class counts:\n{y_train_smote.value_counts()}')

# %% [code]
before_counts = y_train.value_counts().sort_index()
after_counts  = y_train_smote.value_counts().sort_index()

labels = [f'Class {c}' for c in before_counts.index]
x = np.arange(len(labels))

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 5))

bars1 = ax1.bar(x, before_counts.values, color='steelblue', edgecolor='white', width=0.5)
for bar in bars1:
    ax1.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 5,
             str(bar.get_height()), ha='center', va='bottom', fontsize=11)
ax1.set_title('Before SMOTE Imbalance')
ax1.set_xlabel('Class Status (0 = Retained, 1 = Churned)')
ax1.set_ylabel('Sample Count')
ax1.set_xticks(x)
ax1.set_xticklabels(['Non Churned', 'Churned'])
ax1.grid(axis='y', linestyle='--', alpha=0.5)

bars2 = ax2.bar(x, after_counts.values, color='mediumseagreen', edgecolor='white', width=0.5)
for bar in bars2:
    ax2.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 5,
             str(bar.get_height()), ha='center', va='bottom', fontsize=11)
ax2.set_title('After SMOTE Resampling')
ax2.set_xlabel('Class Status (0 = Retained, 1 = Churned)')
ax2.set_ylabel('Sample Count')
ax2.set_xticks(x)
ax2.set_xticklabels(['Non Churned', 'Churned'])
ax2.grid(axis='y', linestyle='--', alpha=0.5)

plt.suptitle('Class Distribution Analysis Before vs After SMOTE Balancing', fontsize=13, fontweight='bold')
plt.tight_layout()
plt.show()
```

Figure A12: Code for Applying SMOTE Class Balancing

```python
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

lr_model = LogisticRegression(random_state=42, max_iter=1000)
lr_model.fit(X_train_smote, y_train_smote)

y_pred_lr = lr_model.predict(X_test_scaled)

print(f'Accuracy: {accuracy_score(y_test, y_pred_lr):.3f}')
print('\nClassification Report:')
print(classification_report(y_test, y_pred_lr, target_names=['Non Churned', 'Churned']))

plt.figure(figsize=(6, 4))
sns.heatmap(confusion_matrix(y_test, y_pred_lr),
            annot=True, fmt='d', cmap='Blues',
            xticklabels=['Non Churned', 'Churned'],
            yticklabels=['Non Churned', 'Churned'])
plt.title('Confusion Matrix - Logistic Regression')
plt.ylabel('Actual')
plt.xlabel('Predicted')
plt.show()
```

Figure A13: Code for Training and Evaluating Logistic Regression

```python
from sklearn.tree import DecisionTreeClassifier

dt_model = DecisionTreeClassifier(random_state=42)
dt_model.fit(X_train_smote, y_train_smote)

y_pred_dt = dt_model.predict(X_test_scaled)

print(f'Accuracy: {accuracy_score(y_test, y_pred_dt):.3f}')
print('\nClassification Report:')
print(classification_report(y_test, y_pred_dt, target_names=['Non Churned', 'Churned']))

plt.figure(figsize=(6, 4))
sns.heatmap(confusion_matrix(y_test, y_pred_dt),
            annot=True, fmt='d', cmap='Blues',
            xticklabels=['Non Churned', 'Churned'],
            yticklabels=['Non Churned', 'Churned'])
plt.title('Confusion Matrix - Decision Tree')
plt.ylabel('Actual')
plt.xlabel('Predicted')
plt.show()
```

Figure A14: Code for Training and Evaluating Decision Tree

```python
from sklearn.ensemble import RandomForestClassifier

rf_model = RandomForestClassifier(random_state=42, n_estimators=100)
rf_model.fit(X_train_smote, y_train_smote)

y_pred_rf = rf_model.predict(X_test_scaled)

print(f'Accuracy: {accuracy_score(y_test, y_pred_rf):.3f}')
print('\nClassification Report:')
print(classification_report(y_test, y_pred_rf, target_names=['Non Churned', 'Churned']))

plt.figure(figsize=(6, 4))
sns.heatmap(confusion_matrix(y_test, y_pred_rf),
            annot=True, fmt='d', cmap='Blues',
            xticklabels=['Non Churned', 'Churned'],
            yticklabels=['Non Churned', 'Churned'])
plt.title('Confusion Matrix - Random Forest')
plt.ylabel('Actual')
plt.xlabel('Predicted')
plt.show()
```

Figure A15: Code for Training and Evaluating Random Forest

```python
from sklearn.svm import SVC

svm_model = SVC(random_state=42, kernel='rbf')
svm_model.fit(X_train_smote, y_train_smote)

y_pred_svm = svm_model.predict(X_test_scaled)

print(f'Accuracy: {accuracy_score(y_test, y_pred_svm):.3f}')
print('\nClassification Report:')
print(classification_report(y_test, y_pred_svm, target_names=['Non Churned', 'Churned']))

plt.figure(figsize=(6, 4))
sns.heatmap(confusion_matrix(y_test, y_pred_svm),
            annot=True, fmt='d', cmap='Blues',
            xticklabels=['Non Churned', 'Churned'],
            yticklabels=['Non Churned', 'Churned'])
plt.title('Confusion Matrix - SVM')
plt.ylabel('Actual')
plt.xlabel('Predicted')
plt.show()
```

Figure A16: Code for Training and Evaluating Support Vector Machine (SVM)

```python
from sklearn.neighbors import KNeighborsClassifier

knn_model = KNeighborsClassifier(n_neighbors=5)
knn_model.fit(X_train_smote, y_train_smote)

y_pred_knn = knn_model.predict(X_test_scaled)

print(f'Accuracy: {accuracy_score(y_test, y_pred_knn):.3f}')
print('\nClassification Report:')
print(classification_report(y_test, y_pred_knn, target_names=['Non Churned', 'Churned']))

plt.figure(figsize=(6, 4))
sns.heatmap(confusion_matrix(y_test, y_pred_knn),
            annot=True, fmt='d', cmap='Blues',
            xticklabels=['Non Churned', 'Churned'],
            yticklabels=['Non Churned', 'Churned'])
plt.title('Confusion Matrix - KNN')
plt.ylabel('Actual')
plt.xlabel('Predicted')
plt.show()
```

Figure A17: Code for Training and Evaluating K-Nearest Neighbors (KNN)

```python
from xgboost import XGBClassifier

xgb_model = XGBClassifier(random_state=42, eval_metric='logloss')
xgb_model.fit(X_train_smote, y_train_smote)

y_pred_xgb = xgb_model.predict(X_test_scaled)

print(f'Accuracy: {accuracy_score(y_test, y_pred_xgb):.3f}')
print('\nClassification Report:')
print(classification_report(y_test, y_pred_xgb, target_names=['Non Churned', 'Churned']))

plt.figure(figsize=(6, 4))
sns.heatmap(confusion_matrix(y_test, y_pred_xgb),
            annot=True, fmt='d', cmap='Blues',
            xticklabels=['Non Churned', 'Churned'],
            yticklabels=['Non Churned', 'Churned'])
plt.title('Confusion Matrix - XGBoost')
plt.ylabel('Actual')
plt.xlabel('Predicted')
plt.show()
```

Figure A18: Code for Training and Evaluating XGBoost

```python
from sklearn.metrics import roc_curve, roc_auc_score

# Place this BEFORE "ROC Curve Comparison"
lr_probs  = lr_model.predict_proba(X_test_scaled)[:, 1]
dt_probs  = dt_model.predict_proba(X_test_scaled)[:, 1]
rf_probs  = rf_model.predict_proba(X_test_scaled)[:, 1]
knn_probs = knn_model.predict_proba(X_test_scaled)[:, 1]
xgb_probs = xgb_model.predict_proba(X_test_scaled)[:, 1]
svm_probs = svm_model.decision_function(X_test_scaled)

models_probs = {
    'Logistic Regression': lr_probs,
    'Decision Tree':       dt_probs,
    'Random Forest':       rf_probs,
    'SVM':                 svm_probs,
    'KNN':                 knn_probs,
    'XGBoost':             xgb_probs
}

# ROC curves — already in your notebook, kept here for reference alongside the metrics table
plt.figure(figsize=(10, 6))
for model_name, probs in models_probs.items():
    fpr, tpr, _ = roc_curve(y_test, probs)
    auc = roc_auc_score(y_test, probs)
    plt.plot(fpr, tpr, label=f'{model_name} (AUC = {auc:.3f})')

plt.plot([0, 1], [0, 1], 'k--', label='Random Classifier')
plt.title('ROC Curve — All Models')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.legend(loc='lower right')
plt.tight_layout()
plt.show()
```

Figure A19: Code for Plotting Multi-Model ROC Curves

```python
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score,
    f1_score, roc_auc_score, roc_curve
)

# Predictions already computed: y_pred_lr, y_pred_dt, y_pred_rf, y_pred_svm, y_pred_knn, y_pred_xgb
# Probabilities already computed: lr_probs, dt_probs, rf_probs, knn_probs, xgb_probs, svm_probs (decision_function)

models_preds = {
    'Logistic Regression': y_pred_lr,
    'Decision Tree':       y_pred_dt,
    'Random Forest':       y_pred_rf,
    'SVM':                 y_pred_svm,
    'KNN':                 y_pred_knn,
    'XGBoost':             y_pred_xgb
}

models_probs = {
    'Logistic Regression': lr_probs,
    'Decision Tree':       dt_probs,
    'Random Forest':       rf_probs,
    'SVM':                 svm_probs,   # decision_function score, fine for AUC
    'KNN':                 knn_probs,
    'XGBoost':             xgb_probs
}

# Build full metrics table — focus on the "Churned" (positive) class for precision/recall/F1
results = []
for name, preds in models_preds.items():
    probs = models_probs[name]
    results.append({
        'Model':     name,
        'Accuracy':  accuracy_score(y_test, preds),
        'Precision': precision_score(y_test, preds, pos_label=1),
        'Recall':    recall_score(y_test, preds, pos_label=1),
        'F1-Score':  f1_score(y_test, preds, pos_label=1),
        'AUC':       roc_auc_score(y_test, probs)
    })

results_df = pd.DataFrame(results).set_index('Model').round(3)
results_df = results_df.sort_values('F1-Score', ascending=False)

print('Model Performance Summary (Churned class = positive):')
print(results_df.to_string())
```

Figure A20: Code for Compiling the Sorted Performance Summary Table

```python
import shap
import matplotlib.pyplot as plt

explainer = shap.TreeExplainer(rf_model)
shap_values = explainer.shap_values(X_test_scaled)

# Future-proof check for both old (list) and new (3D array) SHAP output formats
if isinstance(shap_values, list):
    shap_values_churn = shap_values[1]
elif hasattr(shap_values, "ndim") and shap_values.ndim == 3:
    shap_values_churn = shap_values[:, :, 1]
else:
    shap_values_churn = shap_values

# --- Feature Importance Analysis ---
# (plt.figure() removed as shap builds its own)
shap.summary_plot(shap_values_churn, X_test_scaled,
                  feature_names=X_test_scaled.columns.tolist(),
                  plot_type='bar',
                  show=False)
plt.title('Feature Importance Analysis (Mean |SHAP Value|) – Random Forest')
plt.tight_layout()
plt.show()

# --- SHAP Summary Plot ---
shap.summary_plot(shap_values_churn, X_test_scaled,
                  feature_names=X_test_scaled.columns.tolist(),
                  show=False)
plt.title('SHAP Summary Plot – Random Forest (Churned Class)')
plt.tight_layout()
plt.show()
```

Figure A21: Code for Generating SHAP Feature Importance Bar Plot and Summary Plot

Part B: Execution Outputs & Visualizations

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6000 entries, 0 to 5999
Data columns (total 16 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   Customer_ID                  6000 non-null   object
 1   account_age_months           6000 non-null   int64
 2   avg_order_value              6000 non-null   float64
 3   total_orders                 6000 non-null   int64
 4   days_since_last_purchase     6000 non-null   int64
 5   discount_usage_rate          6000 non-null   float64
 6   return_rate                  6000 non-null   float64
 7   customer_support_tickets     6000 non-null   int64
 8   loyalty_member               6000 non-null   object
 9   browsing_frequency_per_week  6000 non-null   float64
 10  cart_abandonment_rate        6000 non-null   float64
 11  product_review_score_avg     6000 non-null   float64
 12  engagement_score             6000 non-null   float64
 13  satisfaction_score           6000 non-null   float64
 14  price_sensitivity_index      6000 non-null   float64
 15  churned                      6000 non-null   object
dtypes: float64(9), int64(4), object(3)
memory usage: 750.1+ KB
```

Figure A22: Data Structure and Attribute Types Summary Output (df.info())

```
df.describe()
```

	account_age_months	avg_order_value	total_orders	days_since_last_purchase	discount_usage_rate	return_rate	customer_support_tickets	browsing_frequency_per_week
count	6000.000000	6000.000000	6000.00000	6000.000000	6000.000000	6000.000000	6000.000000	6000.000000
mean	30.806667	80.487945	8.56750	29.598167	0.285057	0.071519	0.857667	3.076683
std	17.358115	55.044707	9.88833	29.452645	0.158112	0.065077	0.977702	1.893899
min	1.000000	10.000000	1.00000	0.000000	0.003000	0.000000	0.000000	0.000000
25%	16.000000	44.525000	1.00000	9.000000	0.162000	0.022400	0.000000	1.600000
50%	31.000000	67.225000	5.00000	20.000000	0.265000	0.052800	1.000000	3.000000
75%	46.000000	100.685000	13.00000	41.000000	0.387000	0.102400	1.000000	4.400000
max	60.000000	1006.530000	85.00000	261.000000	0.895000	0.493600	6.000000	10.300000

Figure A23: Data Descriptive Statistics Summary Output (df.describe())

```
Classification Report:
              precision    recall  f1-score   support

 Non Churned       0.99      0.96      0.97      1014
     Churned       0.80      0.95      0.87       186

    accuracy                           0.96      1200
   macro avg       0.90      0.95      0.92      1200
weighted avg       0.96      0.96      0.96      1200
```

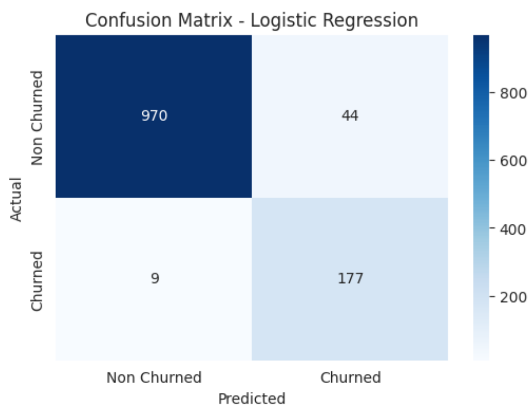

Figure A24: Model Output: Heatmap Confusion Matrix and Classification Report for Logistic Regression

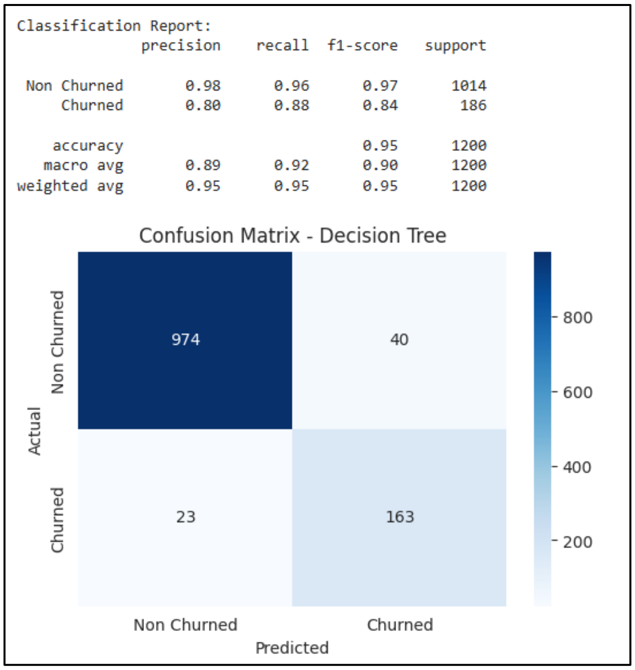

Figure A25: Model Output: Heatmap Confusion Matrix and Classification Report for Decision Tree

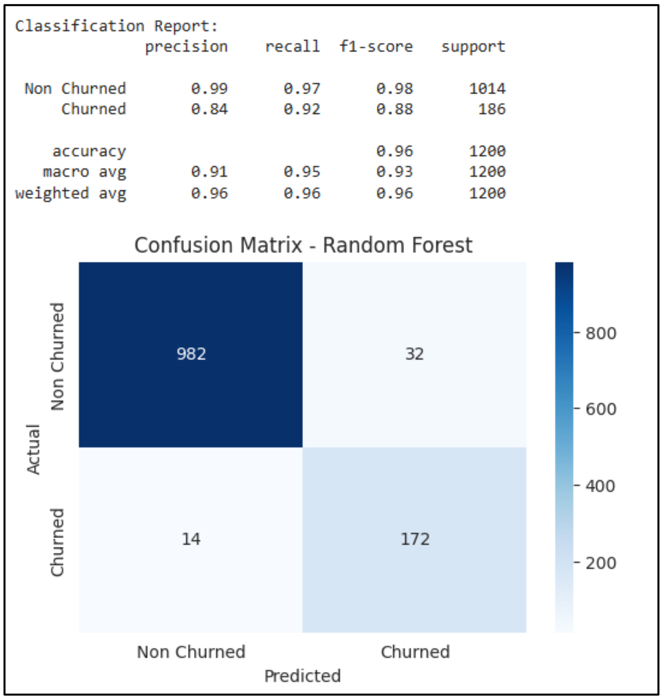

Figure A26: Model Output: Heatmap Confusion Matrix and Classification Report for Random Forest

Classification Report:

	precision	recall	f1-score	support
Non Churned	0.99	0.96	0.97	1014
Churned	0.80	0.93	0.86	186
accuracy			0.95	1200
macro avg	0.90	0.94	0.92	1200
weighted avg	0.96	0.95	0.96	1200

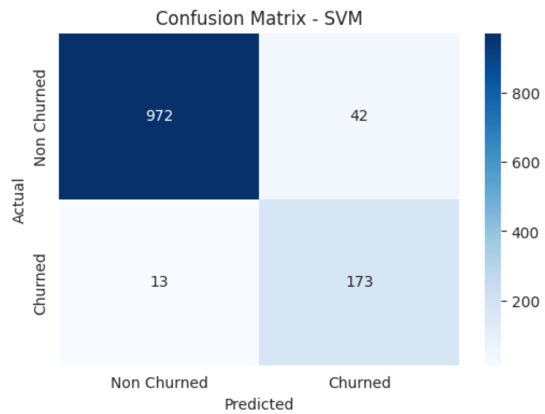

Figure A27: Model Output: Heatmap Confusion Matrix and Classification Report for Support Vector Machine (SVM)

Classification Report:

	precision	recall	f1-score	support
Non Churned	0.99	0.93	0.96	1014
Churned	0.73	0.95	0.82	186
accuracy			0.94	1200
macro avg	0.86	0.94	0.89	1200
weighted avg	0.95	0.94	0.94	1200

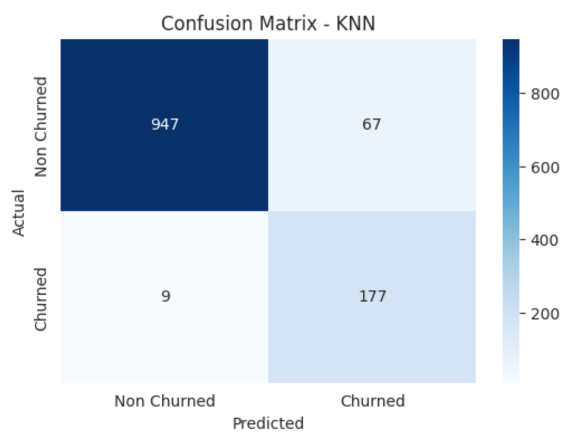

Figure A28: Model Output: Heatmap Confusion Matrix and Classification Report for K-Nearest Neighbors (KNN)

Classification Report:

	precision	recall	f1-score	support
Non Churned	0.98	0.97	0.98	1014
Churned	0.86	0.89	0.88	186
accuracy			0.96	1200
macro avg	0.92	0.93	0.93	1200
weighted avg	0.96	0.96	0.96	1200

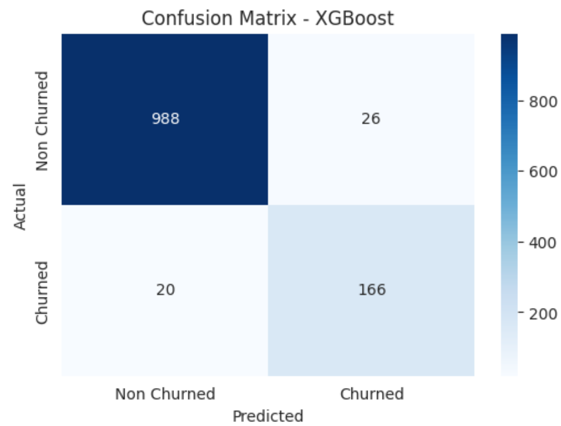

Figure A29: Model Output: Heatmap Confusion Matrix and Classification Report for XGBoost

Model Performance Summary (Churned class = positive):

Model	Accuracy	Precision	Recall	F1-Score	AUC
Random Forest	0.962	0.843	0.925	0.882	0.992
XGBoost	0.962	0.865	0.892	0.878	0.992
Logistic Regression	0.956	0.801	0.952	0.870	0.993
SVM	0.954	0.805	0.930	0.863	0.991
Decision Tree	0.948	0.803	0.876	0.838	0.918
KNN	0.937	0.725	0.952	0.823	0.970

Figure A30: Compiled Multi-Model Performance Summary Table